

論文段落中英對照研讀

版本 C | 藍框注釋對照

A Dynamic Theory of Organizational Knowledge Creation

組織知識創造的動態理論

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Organization Science, Vol. 5, No. 1 (Feb. 1994), pp. 14–37

▶ 閱讀說明

本檔每一段英文原文之後，緊跟一段藍框中文譯文，一段對一段、不拆句不摘要。三張圖表（Figure 1~3）依原文印刷位置就地放入，圖說採中英並列。

本版收錄範圍：引言＋第 2 節（知識創造的基本概念，2.1~2.5）＋第 3 節（組織知識創造過程，3.1~3.5）；第 4 節管理應用未收錄。

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1. 引言 1. Introduction

It is widely observed that the society we live in has been gradually turning into a "knowledge society" (Drucker 1968; Bell 1973; Toffler 1990). The ever increasing importance of knowledge in contemporary society calls for a shift in our thinking concerning innovation in large business organizations—be it technical innovation, product innovation, or strategic or organizational innovation. It raises questions about how organizations process knowledge and, more importantly, how they create new knowledge. Such a shift in general orientation will involve, among other things, a reconceptualization of the organizational knowledge creation processes.

► 中文譯文

我們所處的社會，普遍被觀察到正逐漸轉變為一個「知識社會」（Drucker 1968; Bell 1973; Toffler 1990）。知識在當代社會的重要性與日俱增，這要求我們在思考大型企業組織的創新時做出轉變，無論那是技術創新、產品創新，還是策略或組織上的創新。它帶出了幾個問題：組織如何處理知識，而更重要的是，組織如何創造新知識。這種思考方向上的整體轉變，將牽涉到對「組織知識創造過程」的重新概念化。

The theory of organization has long been dominated by a paradigm that conceptualizes the organization as a system that 'processes' information or 'solves' problems. Central to this paradigm is the assumption that a fundamental task for the organization is how efficiently it can deal with information and decisions in an uncertain environment. This paradigm suggests that the solution lies in the 'input-process-output' sequence of hierarchical information processing. Yet a critical problem with this paradigm follows from its passive and static view of the organization. Information processing is viewed as a problem-solving activity which centers on what is given to the organization—without due consideration of what is created by it.

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長久以來，組織理論一直被一種典範所主導，這個典範把組織理解成一個「處理」資訊或「解決」問題的系統。這個典範的核心是一個假設：組織的根本任務，在於它能多有效率地在不確定的環境中處理資訊與做決策。它認為解方就在於「輸入到處理再到輸出」這種層級式資訊處理的流程。但這個典範有個關鍵問題，來自它對組織所抱持的被動而靜態的觀點。資訊處理被當成一種解決問題的活動，焦點全放在「外界給了組織什麼」，卻沒有好好考慮「組織自己創造了什麼」。

Any organization that dynamically deals with a changing environment ought not only to process information efficiently but also create information and knowledge. Analyzing the organization in terms of its design and capability to process information imposed by the environment no doubt constitutes an important approach to interpreting certain aspects of organizational activities. However, it can be argued that the organization's interaction with its environment, together with the means by which it creates and distributes information and knowledge, are more important when it comes to building an active and dynamic understanding of the organization. For example, innovation, which is a key form of organizational knowledge creation, cannot be explained sufficiently in terms of information processing or problem solving. Innovation can be better understood as a process in which the organization creates and defines problems and then actively develops new knowledge to solve them. Also, innovation produced by one part of the organization in turn creates a stream of related information and knowledge, which might then trigger changes in the organization's wider knowledge systems. Such a sequence of innovation suggests that the organization should be studied from the viewpoint of how it creates information and knowledge, rather than with regard to how it processes these entities.

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任何要動態地因應變動環境的組織，不該只是有效率地處理資訊，還應該創造資訊與知識。從「組織的設計、以及它處理環境所加諸資訊的能力」來分析組織，無疑是詮釋組織活動某些面向的重要途徑。但可以這樣主張：當我們要建立對組織「主動而動態」的理解時，組織與環境之間的互動、以及它創造與散布資訊和知識的方式，其實更為重要。舉例來說，創新是組織知識創造的一種關鍵形式，它沒辦法只用資訊處理或問題解決來充分解釋。創新更應該被理解成這樣一個

過程：組織在其中創造並界定問題，然後主動發展新知識去解決這些問題。而且，組織某一部分所產生的創新，又會反過來帶出一連串相關的資訊與知識，進而可能觸發組織更廣泛知識系統的改變。這樣的創新連鎖說明了：研究組織時，應該從「它如何創造資訊與知識」的角度切入，而不是只看「它如何處理這些東西」。

The goal of this paper is to develop the essential elements of a theory of organizational knowledge creation. In the sections which follow, the basic concepts and models of the theory of organizational knowledge creation are presented. Based on this foundation, the dynamics of the organizational knowledge creation process are examined and practical models are advanced for managing the process more effectively.

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本文的目標，是發展出一套「組織知識創造理論」的核心要素。在接下來的章節裡，會先說明這套理論的基本概念與模型。在這個基礎之上，再進一步檢視組織知識創造過程的動態，並提出可實際操作的模型，以更有效地管理這個過程。

2. 組織知識創造的基本概念與模型 2. Basic Concepts and Models of Organizational Knowledge Creation

The following subsections explore some basic constructs of the theory of organizational knowledge creation. They begin by discussing the nature of information and knowledge and then draw a distinction between "tacit" and "explicit" knowledge. This distinction represents what could be described as the epistemological dimension to organizational knowledge creation. It embraces a continual dialogue between explicit and tacit knowledge which drives the creation of new ideas and concepts.

► 中文譯文

以下幾個小節，會探討組織知識創造理論的一些基本構念。先從討論資訊與知識的本質開始，接著區分「隱性」知識與「顯性」知識。這個區分，代表的是組織知識創造的「認識論維度」。它包含顯性知識與隱性知識之間持續不斷的對話，而正是這個對話，推動了新想法與新概念的產生。

Although ideas are formed in the minds of individuals, interaction between individuals typically plays a critical role in developing these ideas. That is to say, "communities of interaction" contribute to the amplification and development of new knowledge. While these communities might span departmental or indeed organizational boundaries, the point to note is that they define a further dimension to organizational knowledge creation, which is associated with the extent of social interaction between individuals that share and develop knowledge. This is referred to as the "ontological" dimension of knowledge creation.

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雖然想法是在個人的腦中形成的，但個人與個人之間的互動，通常在發展這些想法上扮演關鍵角色。也就是說，「互動社群」對新知識的擴大與發展是有貢獻的。這些社群可能跨越部門、甚至跨越組織的邊界，但值得注意的是，它們界定了組織知識創造的另一個維度，這個維度跟「分享並發展知識的個人之間，社會互動的程度」有關。這就是知識創造的「本體論維度」。

Following a consideration of the two dimensions of knowledge creation, some attention is given to the role of individuals and, more specifically, to their "commitment" to the knowledge creating process. This covers aspects of their "intention," the role of autonomy, and the effects of fluctuations or discontinuities in the organization and its environment.

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在討論完知識創造的這兩個維度之後，接著會把焦點放在個人的角色上，更具體地說，是放在個人對知識創造過程的「承諾」上。這部分涵蓋了他們的「意圖」、自主性所扮演的角色，以及組織及其環境中的波動或斷裂所帶來的影響。

Next, a "spiral" model of knowledge creation is proposed which shows the relationship between the epistemological and ontological dimensions of knowledge creation. This spiral illustrates the creation of a new concept in terms of a continual dialogue between tacit and explicit knowledge. As the concept resonates around an expanding community of individuals, it is developed and clarified. Gradually, concepts which are thought to be of value obtain a wider currency and become crystalized. This description of the spiral model is followed by some observations about how to support the practical management of organizational knowledge creation.

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接下來，會提出一個知識創造的「螺旋」模型，用來呈現知識創造的認識論維度與本體論維度之間的關係。這個螺旋，是用「隱性知識與顯性知識之間持續對話」的方式，來說明一個新概念是如何產生的。當這個概念在一個不斷擴大的個人社群裡產生共鳴，它就會被發展、被釐清。漸漸地，那些被認為有價值的概念，會取得更廣泛的流通，並且結晶成形。在描述完這個螺旋模型之後，會再提出一些觀察，談談如何支援組織知識創造的實務管理。

2.1 知識與資訊 2.1 Knowledge and Information

Knowledge is a multifaceted concept with multilayered meanings. The history of philosophy since the classical Greek period can be regarded as a never-ending search for the meaning of knowledge. This paper follows traditional epistemology and adopts a definition of knowledge as "justified true belief." It should be noted, however, that while the arguments of traditional epistemology focus on "truthfulness" as the essential attribute of knowledge, for present purposes it is important to consider knowledge as a personal "belief," and emphasize the importance of the "justification" of knowledge. This difference introduces another critical distinction between the view of knowledge of traditional epistemology and that of the theory of knowledge creation. While the former naturally emphasizes the absolute, static, and nonhuman nature of knowledge, typically expressed in propositional forms in formal logic, the latter sees knowledge as a dynamic human process of justifying personal beliefs as part of an aspiration for the "truth."

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知識是一個多面向、意義多層次的概念。自古希臘時期以來的哲學史，可以看成是對「知識意義」一場永無止境的探尋。本文沿用傳統的認識論，採用「知識是經過證成的真實信念」這個定義。不過要留意的是，傳統認識論的論述把「真實性」當作知識的本質屬性，但就本文的目的而言，更重要的是把知識看成一種個人的「信念」，並強調知識「證成」過程的重要性。這個差異，引出了傳統認識論的知識觀與知識創造理論的知識觀之間另一個關鍵區別：前者自然會強調知識那種絕對的、靜態的、非人性的本質，通常以形式邏輯中的命題形式來表達；後者則把知識看成一個動態的人性過程，是人為了追求「真理」而不斷證成個人信念的過程。

Although the terms "information" and "knowledge" are often used interchangeably, there is a clear distinction between information and knowledge. According to Machlup (1983), information is a flow of messages or meanings which might add to, restructure or change knowledge. Dretske (1981) offers some useful definitions. In his words:

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雖然「資訊」與「知識」這兩個詞常常被交替使用，但資訊與知識之間有著清楚的區別。根據 Machlup (1983) 的說法，資訊是訊息或意義的一股流動，它可能增添、重構或改變知識。Dretske (1981) 則提供了一些有用的定義。用他的話說：

Information is that commodity capable of yielding knowledge, and what information a signal carries is what we can learn from it (Dretske 1981, p. 44). Knowledge is identified with information-produced (or sustained) belief, but the information a person receives is relative to what he or she already knows about the possibilities at the source (ibid, p. 86).

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資訊是一種能夠產出知識的東西，而一個訊號所承載的資訊，就是我們能從它身上學到的東西 (Dretske 1981, p. 44)。知識被認為「由資訊產生（或維繫）的信念」，但一個人所接收

到的資訊，是相對於他或她對訊息源頭那些可能性「原本已經知道多少」而定的（同上，p. 86）。

In short, information is a flow of messages, while knowledge is created and organized by the very flow of information, anchored on the commitment and beliefs of its holder. This understanding emphasizes an essential aspect of knowledge that relates to human action.

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簡而言之，資訊是訊息的流動，而知識則是由這股資訊流動所創造與組織起來的，並且錨定在「知識持有者的承諾與信念」之上。這樣的理解，凸顯了知識一個與人類行動有關的本質面向。

The importance of knowledge related to action has been recognized in the area of artificial intelligence. For example, Gruber (1989) addresses the subject of an expert's "strategic knowledge" as that which directly guides his action, and attempts to develop the tools to acquire it. Since the 1980s, the development of cognitive science has been based on a serious reflection on behaviorist psychology's neglect of such traditional questions as, 'Why do human beings act in a certain way?', which was a central issue for so-called "folk psychology" (Stich 1986). Searle's discussion on the "speech act" also points out a close relationship between language and human action in terms of the "intention" and "commitment" of speakers (Searle 1969). In sum, as a fundamental basis for the theory of organizational creation of knowledge, it can be argued that attention should be focused on the active, subjective nature of knowledge represented by such terms as "belief" and "commitment" that are deeply rooted in the value systems of individuals.

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知識與行動有關，這一點的重要性，在人工智慧領域已經獲得認可。舉例來說，Gruber（1989）探討了專家的「策略性知識」，也就是那種直接引導其行動的知識，並嘗試發展出擷取這種知識的工具。自 1980 年代以來，認知科學的發展，建立在對行為主義心理學的一個嚴肅反省之上：行為主義忽略了像「人為什麼會以某種特定方式行動？」這類傳統問題，而這正是所謂「常民心理學」的核心議題（Stich 1986）。Searle 對「言語行為」的討論，也從說話者的「意圖」與「承諾」這個角度，指出了語言與人類行動之間有著緊密的關係（Searle 1969）。總的來說，作為組織知識創造理論的根本基礎，可以這樣主張：我們應該把注意力放在知識那種主動的、主觀的本質上，也就是「信念」與「承諾」這類深植於個人價值體系中的詞語所代表的東西。

The analysis of knowledge and information does not stop at this point. Information is a necessary medium or material for initiating and formalizing knowledge and can be viewed from "syntactic" and "semantic" perspectives. The syntactic aspect of information is illustrated by Shannon's analysis of the volume of information which is measured without regard to its meaning or value. A telephone bill, for example, is not calculated on the basis of the content of a conversation but according to the duration of time and the distance involved. Shannon said that the semantic aspects of communication, which center on the meaning of information, are irrelevant to the engineering problem (Shannon and Weaver 1949). A genuine theory of information would be a theory about the content of our messages, not a theory about the form in which this content is embodied (Dretske 1981).

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對知識與資訊的分析，並不止步於此。資訊是「啟動並形式化知識」的必要媒介或材料，而它可以從「語法」與「語意」兩個角度來看。資訊的語法面向，可以用 Shannon 對「資訊量」的分析來說明：這種測量並不考慮資訊的意義或價值。舉例來說，電話帳單並不是依據對話的內容來計算，而是依據通話的時間長度與距離。Shannon 說過，溝通中那些以「資訊意義」為核心的語意面向，跟工程上的問題無關（Shannon and Weaver 1949）。然而一個真正的資訊理論，應該是關於「我們訊息的內容」的理論，而不是關於「這些內容被承載的形式」的理論（Dretske 1981）。

In terms of creating knowledge, the semantic aspect of information is more relevant as it focuses on conveyed meaning. The syntactic aspect does not capture the importance of information in the knowledge creation process. Therefore, any

preoccupation with the formal definition will tend to lead to a disproportionate emphasis on the role of information processing, which is insensitive to the creation of organizational knowledge out of the chaotic, equivocal state of information. Information, seen from the semantic standpoint, literally means that it contains new meaning. As Bateson (1979, p. 5) put it, "information consists of differences that make a difference." This insight provides a new point of view for interpreting events that make previously invisible connections or ideas obvious or shed light on unexpected connections (Miyazaki and Ueno 1985). For the purposes of building a theory of knowledge creation, it is important to concentrate on the semantic aspects of information.

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就創造知識而言，資訊的語意面向更為相關，因為它聚焦在「被傳達的意義」上。語法面向則捕捉不到資訊在知識創造過程中的重要性。因此，一旦過度執著於形式上的定義，往往會導致對「資訊處理」這個角色不成比例的強調，而這種強調，對「從混沌、模稜兩可的資訊狀態中創造出組織知識」這件事是無感的。從語意的立場來看，資訊在字面上的意思，就是它含有新的意義。如同 Bateson (1979, p. 5) 所說：「資訊由那些『造成差別的差異』所構成。」這個洞見提供了一個新的視角，去詮釋那些把過去看不見的連結或想法變得明顯、或照亮意料之外連結的事件 (Miyazaki and Ueno 1985)。為了建立一套知識創造理論，把焦點集中在資訊的語意面向上，是很重要的。

2.2 知識創造的兩個維度 2.2 Two Dimensions of Knowledge Creation

Although a great deal has been written about the importance of knowledge in management, relatively little attention has been paid to how knowledge is created and how the knowledge creation process can be managed. One dimension of this knowledge creation process can be drawn from a distinction between two types of knowledge—"tacit knowledge" and "explicit knowledge." As Michael Polanyi (1966, p. 4) put it, "We can know more than we can tell". Knowledge that can be expressed in words and numbers only represents the tip of the iceberg of the entire body of possible knowledge. Polanyi classified human knowledge into two categories. "Explicit" or codified knowledge refers to knowledge that is transmittable in formal, systematic language. On the other hand, "tacit" knowledge has a personal quality, which makes it hard to formalize and communicate. Tacit knowledge is deeply rooted in action, commitment, and involvement in a specific context. In Polanyi's words, it "indwells" in a comprehensive cognizance of the human mind and body.

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雖然關於知識在管理上的重要性，已經有大量的論述，但相對來說，很少有人關注「知識是如何被創造的」以及「知識創造過程能如何被管理」。這個知識創造過程的其中一個維度，可以從兩種知識的區分中導出，那就是「隱性知識」與「顯性知識」。如同 Michael Polanyi (1966, p. 4) 所說：「我們所能知道的，比我們所能說出來的還要多。」能夠用文字與數字表達出來的知識，只不過是「所有可能知識」這整座冰山的一角。Polanyi 把人類知識分成兩類。「顯性」或已編碼的知識，指的是可以透過正式、有系統的語言來傳遞的知識。另一方面，「隱性」知識則帶有一種個人特質，使它很難被形式化、很難被溝通傳達。隱性知識深植於行動、承諾，以及在特定情境中的投入。用 Polanyi 的話來說，它「內居」於人類身心的整體認知之中。

While Polanyi articulates the contents of tacit knowledge in a philosophical context, it is also possible to expand his idea in a more practical direction. Tacit knowledge involves both cognitive and technical elements. The cognitive elements center on what Johnson-Laird (1983) called "mental models" in which human beings form working models of the world by creating and manipulating analogies in their minds. These working models include schemata, paradigms, beliefs, and viewpoints that provide "perspectives" that help individuals to perceive and define their world. By contrast, the technical element of tacit knowledge covers concrete know-how, crafts, and skills that apply to specific contexts. It is important to note here that the cognitive element of tacit knowledge refers to an individual's images of reality and visions for the future, that is to say, what is and what ought to be. As will be discussed later, the articulation of tacit perspectives—in a kind of "mobilization" process—is a key factor in the creation of new knowledge.

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Polanyi 是在哲學的脈絡下闡述隱性知識的內涵，但我們也可以把他的想法往更實務的方向延伸。

隱性知識同時包含「認知」與「技術」兩種元素。認知元素的核心，是 Johnson-Laird (1983) 所說的「心智模型」：人類在腦中創造並操作各種類比，藉此形成對世界的運作模型。這些運作模型包括基模、典範、信念與觀點，它們提供了一種「視角」，幫助個人去感知並界定自己的世界。相對地，隱性知識的技術元素，涵蓋的是適用於特定情境的具體訣竅、工藝與技能。這裡要特別注意：隱性知識的認知元素，指的是個人對現實的種種圖像、以及對未來的種種願景，也就是「事情現在是什麼樣子」與「事情應該是什麼樣子」。如同後面會討論到的，把隱性的視角清楚表述出來，這個有點像「動員」的過程，正是創造新知識的一個關鍵因素。

Tacit knowledge is a continuous activity of knowing and embodies what Bateson (1973) has referred to as an "analogue" quality. In this context, communication between individuals may be seen as an analogue process that aims to share tacit knowledge to build mutual understanding. This understanding involves a kind of "parallel processing" of the complexities of current issues, as the different dimensions of a problem are processed simultaneously. By contrast, explicit knowledge is discrete or "digital." It is captured in records of the past such as libraries, archives, and databases and is assessed on a sequential basis.

► 中文譯文

隱性知識是一種持續不斷的「知」的活動，它體現了 Bateson (1973) 所說的那種「類比」特質。在這個意義下，個人與個人之間的溝通，可以看成一種類比式的過程，目的是分享隱性知識、建立彼此的理解。這種理解牽涉到對當前議題複雜性的一種「平行處理」，因為一個問題的不同面向是被同時處理的。相對地，顯性知識則是離散的，或者說是「數位的」。它被捕捉在過去的紀錄裡，例如圖書館、檔案庫與資料庫，並且是以循序的方式被取用評估。

本體論維度：社會互動的層級 The Ontological Dimension: The Level of Social Interaction

At a fundamental level, knowledge is created by individuals. An organization cannot create knowledge without individuals. The organization supports creative individuals or provides a context for such individuals to create knowledge. Organizational knowledge creation, therefore, should be understood in terms of a process that "organizationally" amplifies the knowledge created by individuals, and crystallizes it as a part of the knowledge network of organization.

► 中文譯文

在最根本的層次上，知識是由個人創造的。一個組織若沒有個人，就無法創造知識。組織所做的，是支持有創造力的個人，或者提供一個情境，讓這些個人去創造知識。因此，組織知識創造應該被理解成這樣一個過程：它把個人所創造的知識，以「組織的方式」加以擴大，並使其結晶，成為組織知識網絡的一部分。

In this line, it is possible to distinguish several levels of social interaction at which the knowledge created by an individual is transformed and legitimized. In the first instance, an informal community of social interaction provides an immediate forum for nurturing the emergent property of knowledge at each level and developing new ideas. Since this informal community might span organizational boundaries—for example, to include suppliers or customers—it is important that the organization is able to integrate appropriate aspects of emerging knowledge into its strategic development. Thus, the potential contribution of informal groups to organizational knowledge creation should be related to more formal notions of a hierarchical structure. If this is done effectively, new knowledge associated with more advantageous organizational processes or technologies will be able to gain a broader currency within the organization.

► 中文譯文

順著這個思路，我們可以區分出好幾個社會互動的層級，個人所創造的知識，就是在這些層級上被轉化、被賦予正當性。首先，一個非正式的社會互動社群，提供了一個即時的場域，去孕育各層級知識那種「逐漸浮現」的特質，並發展新的想法。由於這種非正式社群可能跨越組織的邊界，例如把供應商或顧客也納進來，因此組織能不能把這些正在浮現的知識中合適的部分，整合進自己的策略發展，就變得很重要。也因此，非正式群體對組織知識創造的潛在貢獻，應該要跟比較

正式的層級結構概念連結起來。如果這件事做得有效，那些跟更有利的組織流程或技術相關的新知識，就能在組織內部取得更廣泛的流通。

In addition to the creation of knowledge within an organization, it is also possible that there will be formal provisions to build knowledge at an interorganizational level. This might occur if informal communities of interaction, that span the link between customers, suppliers, distributors, and even competitors, are put on a more formal basis, for example, through the formation of alliances or outsourcing.

► 中文譯文

除了在組織「內部」創造知識之外，也有可能出現一些正式的安排，在「組織之間」的層級上建構知識。這種情況可能發生在：那些橫跨顧客、供應商、經銷商、甚至競爭對手之間連結的非正式互動社群，被放到一個比較正式的基礎上運作，例如透過結成聯盟或委外的方式。

2.3 知識主體的承諾：意圖、自主與波動 2.3 Commitment on the Part of the Knowledge Subject: Intention, Autonomy, and Fluctuation

The prime movers in the process of organizational knowledge creation are the individual members of an organization. Individuals are continuously committed to recreating the world in accordance with their own perspectives. As Polanyi noted, "commitment" underlies human knowledge creating activities. Thus, commitment is one of the most important components for promoting the formation of new knowledge within an organization. There are three basic factors that induce individual commitment in an organizational setting: "intention," and "autonomy," and a certain level of environmental "fluctuation."

► 中文譯文

在組織知識創造的過程裡，真正的推動者是組織裡的個別成員。個人持續地投入，依照自己的視角去重新創造這個世界。如同 Polanyi 所指出的，「承諾」是人類知識創造活動的底層基礎。因此，承諾是促進組織內部新知識形成最重要的構成要素之一。在組織的情境裡，有三個基本因素會引發個人的承諾：「意圖」、「自主」，以及某種程度的環境「波動」。

意圖 Intention

Intention is concerned with how individuals form their approach to the world and try to make sense of their environment. It is not simply a state of mind, but rather what might be called an action-oriented concept. Edmund Husserl (1968) called this attitude on the part of the subject "intentionality." He denied the existence of "consciousness" per se, which was generally assumed by psychologists in 19th century, and argued that consciousness arises when a subject pays attention to an object. In other words, any consciousness is a 'consciousness of something.' It arises, endures, and disappears with a subject's commitment to an object.

► 中文譯文

意圖關乎的是：個人如何形成他們面對世界的方式，以及如何試著去理解自己所處的環境。它不只是一種心理狀態，反而更應該被稱為一種「以行動為導向」的概念。Edmund Husserl (1968) 把主體這種態度稱為「意向性」。他否認有「意識」這個東西本身的存在，也就是十九世紀心理學家普遍假定的那種意識，並主張意識是在「一個主體把注意力投向某個對象」時才生起的。換句話說，任何意識都是「對某個東西的意識」。意識會隨著一個主體對某個對象的承諾，而生起、持續、然後消失。

Eigen (1971) argued, in his evolutionary theory, that evolution involves the process of acquiring environmental information for better adaptation. Eigen insisted that the degree of meaningfulness of information, or a value parameter, needs to be introduced to explain this system. Human beings, as organic systems, derive meaning from the environment which is based on their ultimate pursuit of survival (Shimizu 1978). Man cannot grasp the meaning of information about his environment without some frame of value judgment.

► 中文譯文

Eigen (1971) 在他的演化理論中主張，演化牽涉到「为了更好地適應而取得環境資訊」的過程。Eigen 堅持，要解釋這個系統，就必須引進「資訊的意義程度」這個概念，或者說一個價值參數。人類作為一種有機系統，是從環境中汲取意義的，而這種汲取，奠基於他們對「生存」的終極追求 (Shimizu 1978)。人若是少了某種價值判斷的框架，就無法掌握關於自身環境的資訊所代表的意義。

The meaning of information differs according to what a particular system aims to do (manifest purpose or problem consciousness) and the broader environment in which that system exists (context). It is more concerned with the system's future aspirations than its current state. Weick (1979) explains this "self-fulfilling prophecy" of a system as the "enactment" of the environment, which may be a projection of its strong will for self-actualization. While mechanistic information-processing models treat the mind as a fixed capacity device for converting meaningless information into conscious perception, in reality cognition is the activity of knowing and understanding as it occurs in the context of purposeful activity (Neisser 1976). Intention becomes apparent against this background. Without intention, it would be impossible to judge the value of the information or knowledge perceived or created. "The intentionality of the mind not only creates the possibility of meaning, but also limits its form" (Searle 1983, p. 166).

► 中文譯文

資訊的意義，會依據兩件事而有所不同：一是某個特定系統想要做什麼（它外顯的目的，或問題意識），二是這個系統所存在的更大環境（脈絡）。它更關乎的是系統對未來的期望，而不是它當下的狀態。Weick (1979) 把系統這種「自我實現的預言」解釋成對環境的「促動」，而這可能是系統那股強烈「自我實現意志」的投射。機械式的資訊處理模型，把心智當成一個固定容量的裝置，負責把無意義的資訊轉換成有意識的知覺；但實際上，認知是一種「知與理解」的活動，它發生在「有目的的活動」這個脈絡之中 (Neisser 1976)。意圖，正是在這樣的背景下才變得清晰可見。沒有意圖，就不可能去判斷所感知到、或所創造出的資訊或知識的價值。「心智的意向性，不只創造了意義的可能，同時也限制了意義的形式」 (Searle 1983, p. 166)。

自主 Autonomy

The principle of autonomy can be applied at the individual, group, and organizational levels—either separately or all together. However, the individual is a convenient starting point for analysis. Individuals within the organization may have different intentions. Every individual has his or her own personality. By allowing people to act autonomously, the organization may increase the possibility of introducing unexpected opportunities of the type that are sometimes associated with the so-called "garbage can" metaphor (Cohen et al. 1972). From the standpoint of creating knowledge, such an organization is more likely to maintain greater flexibility in acquiring, relating, and interpreting information. In a system where the autonomy of individuals is assured, or where only "minimum critical specification" (Morgan 1986) is intended, it is possible to establish a basis for self-organization.

► 中文譯文

自主原則可以套用在個人、群體與組織三個層級上，可以分別套用，也可以全部一起套用。不過，個人是個方便的分析起點。組織裡的不同個人，可能各自懷著不同的意圖。每一個人都有他或她自己的個性。透過容許人們自主行動，組織就可能提高引進「意料之外機會」的可能性，這類機會有時會跟所謂的「垃圾桶」隱喻連在一起 (Cohen et al. 1972)。從創造知識的立場來看，這樣的組織比較可能在取得、連結與詮釋資訊時，維持更大的彈性。在一個「個人自主性受到保障」、或者「只設定最低限度關鍵規範」 (Morgan 1986) 的系統裡，就有可能為自我組織建立起基礎。

Individual autonomy widens the possibility that individuals will motivate themselves to form new knowledge. Self-motivation based on deep emotions, for example, in the poet's creation of new expressions, serves as a driving force for the creation of metaphors. A sense of purpose and autonomy becomes important as an organizational context. Purpose serves as the basis of conceptualization. Autonomy gives individuals freedom to absorb knowledge.

► 中文譯文

個人的自主，會拓寬「個人自我激勵去形成新知識」的可能性。奠基於深層情感的自我激勵，例如詩人在創造新的表達方式時所展現的那種，正是創造隱喻的一股驅動力。作為一種組織脈絡，「目的感」與「自主」變得很重要。目的感是概念化的基礎，而自主則賦予個人吸收知識的自由。

波動 Fluctuation

Even though intention is internal to the individual, knowledge creation at the individual level involves continuous interaction with the external world. In this connection, chaos or discontinuity can generate new patterns of interaction between individuals and their environment. Individuals recreate their own systems of knowledge to take account of ambiguity, redundancy, noise, or randomness generated from the organization and its environment. These fluctuations differ from complete disorder and are characterized by "order without recursiveness"—which represents an order where the pattern is hard to predict in the beginning (Gleick 1987).

► 中文譯文

即使意圖是個人內在的東西，個人層次的知識創造，仍然牽涉到跟外在世界持續不斷的互動。在這個關聯上，混沌或斷裂能夠在個人與其環境之間，產生新的互動模式。個人會重新創造自己的知識系統，把組織及其環境所產生的模糊、冗餘、雜訊或隨機納入考量。這些波動跟「完全的失序」不同，它的特徵是一種「沒有遞迴性的秩序」，也就是一種「一開始很難預測其模式」的秩序（Gleick 1987）。

Winograd and Flores (1986) emphasize the role of periodic "breakdowns" in human perception. Breakdown refers to the interruption of an individual's habitual, comfortable 'state-of-being.' When breakdowns occur, individuals question the value of habits and routine tools, which might lead to a realignment of commitments. Environmental fluctuation often triggers this breakdown. When people face such a breakdown or contradiction, they have an opportunity to reconsider their fundamental thinking and perspectives. In other words, they begin to question the validity of basic attitudes toward the world. This process necessarily involves deep personal commitment by the individual and is similar in context to Piaget's (1974) observations about the importance of the role of contradiction in the interaction between the subject and its environment in such a way that the subject forms perceptions through behavior.

► 中文譯文

Winograd 與 Flores (1986) 強調了週期性「崩解」在人類知覺中的作用。崩解指的是一個人習慣性、舒適的「存在狀態」被打斷。當崩解發生時，個人會去質疑習慣與例行工具的價值，而這可能導致承諾的重新調整。環境的波動，常常會觸發這種崩解。當人們面對這樣的崩解或矛盾時，他們就有機會重新思考自己根本的思維與視角。換句話說，他們開始去質疑「自己面對世界的基本態度」是否站得住腳。這個過程必然牽涉到個人深層的承諾，而它在脈絡上，跟 Piaget (1974) 的觀察很相似：Piaget 談的是「矛盾」在主體與其環境互動中的重要作用，也就是主體透過行為來形成知覺的那種互動。

2.4 知識轉換與知識的螺旋 2.4 Knowledge Conversion and the Spiral of Knowledge

It is now possible to bring together the epistemological and ontological dimensions of knowledge creation to form a "spiral" model for the processes involved. This involves identifying four different patterns of interaction between tacit and explicit knowledge. These patterns represent ways in which existing knowledge can be "converted" into new knowledge. Social interaction between individuals then provides an ontological dimension to the expansion of knowledge.

► 中文譯文

現在，我們可以把知識創造的認識論維度與本體論維度結合起來，為其中涉及的種種過程，形成一個「螺旋」模型。這牽涉到去辨識出隱性知識與顯性知識之間四種不同的互動模式。這些模式代表的是「既有知識能被『轉換』成新知識」的種種方式。而個人與個人之間的社會互動，則為知識的擴展提供了一個本體論的維度。

The idea of "knowledge conversion" may be traced from Anderson's ACT model (Anderson 1983) developed in cognitive psychology. In the ACT model, knowledge is divided into "declarative knowledge" (actual knowledge) that is expressed in the form of propositions and "procedural knowledge" (methodological knowledge) which is used in such activities as remembering how to ride a bicycle or play the piano. In the context of the present discussion, the former might approximate to explicit knowledge and the latter to tacit knowledge. Anderson's model hypothesizes that declarative knowledge has to be transformed into procedural knowledge in order for cognitive skills to develop. This hypothesis is consistent with Ryle's classification (1949) of knowledge into categories of knowing that something "exists" and knowing "how" it operates. Anderson's categorization can be regarded as a more sophisticated version of Ryle's classification. One limitation of the ACT model is the hypothesis that transformation of knowledge is unidirectional and only involves transformations from declarative to procedural knowledge, while it can be argued that transformation is bidirectional. This may be because the ACT model is more concerned with maturation than with the creation of knowledge.

► 中文譯文

「知識轉換」這個概念，可以追溯到認知心理學中發展出來的 Anderson 的 ACT 模型 (Anderson 1983)。在 ACT 模型裡，知識被分成兩種：一種是「陳述性知識」（實際的知識），以命題的形式來表達；另一種是「程序性知識」（方法性的知識），用在像是「記得怎麼騎腳踏車或彈鋼琴」這類活動上。放在本文的討論脈絡裡，前者大致對應到顯性知識，後者則對應到隱性知識。Anderson 的模型假設：要讓認知技能得以發展，陳述性知識必須被轉化成程序性知識。這個假設，跟 Ryle (1949) 把知識分成「知道某事物『存在』」與「知道它『如何』運作」兩類的分類是一致的。Anderson 的分類，可以看成是 Ryle 分類的一個更精緻的版本。ACT 模型的一個限制在於：它假設知識的轉化是單向的，只牽涉到從陳述性知識到程序性知識的轉化；然而我們其實可以主張，這種轉化是雙向的。會這樣，或許是因為 ACT 模型更關心的是知識的「成熟」，而不是知識的「創造」。

知識轉換的四種模式 Four Modes of Knowledge Conversion

The assumption that knowledge is created through conversion between tacit and explicit knowledge allows us to postulate four different "modes" of knowledge conversion: (1) from tacit knowledge to tacit knowledge, (2) from explicit knowledge to explicit knowledge, (3) from tacit knowledge to explicit knowledge, and (4) from explicit knowledge to tacit knowledge.

► 中文譯文

「知識是透過隱性知識與顯性知識之間的轉換而被創造出來」這個假設，讓我們可以假定出四種不同的知識轉換「模式」：（1）從隱性知識到隱性知識，（2）從顯性知識到顯性知識，（3）從隱性知識到顯性知識，以及（4）從顯性知識到隱性知識。

First, there is a mode of knowledge conversion that enables us to convert tacit knowledge through interaction between individuals. One important point to note here is that an individual can acquire tacit knowledge without language. Apprentices work with their mentors and learn craftsmanship not through language but by observation, imitation, and practice. In a business setting, on-the-job training (OJT) uses the same principle. The key to acquiring tacit knowledge is experience. Without some form of shared experience, it is extremely difficult for people to share each others' thinking processes. The mere transfer of information will often make little sense if it is abstracted from embedded emotions and nuanced contexts that are associated with shared experiences. This process of creating tacit knowledge through shared experience will be called "socialization."

► 中文譯文

首先，有一種知識轉換的模式，讓我們能夠透過個人與個人之間的互動，去轉換隱性知識。這裡有一個重點要注意：個人是可以在不靠語言的情況下取得隱性知識的。學徒跟著師傅工作，學會手藝，靠的不是語言，而是觀察、模仿與實作。在企業的情境裡，在職訓練（OJT）用的也是同樣的原理。取得隱性知識的關鍵，在於經驗。少了某種形式的共享經驗，人們要分享彼此的思考過程是極為困難的。如果只是把資訊單純地傳遞出去，卻把它從那些「鑲嵌在共享經驗中的情感

與細膩脈絡」抽離出來，那這樣的傳遞往往沒什麼意義。這種「透過共享經驗來創造隱性知識」的過程，會被稱為「社會化」。

The second mode of knowledge conversion involves the use of social processes to combine different bodies of explicit knowledge held by individuals. Individuals exchange and combine knowledge through such exchange mechanisms as meetings and telephone conversations. The reconfiguring of existing information through the sorting, adding, recategorizing, and recontextualizing of explicit knowledge can lead to new knowledge. Modern computer systems provide a graphic example. This process of creating explicit knowledge from explicit knowledge is referred to as "combination."

► 中文譯文

第二種知識轉換的模式，牽涉到運用社會性的過程，去結合個人手上不同的顯性知識。個人會透過像開會、講電話這類交換機制，去交換並結合知識。對既有資訊進行重組，也就是對顯性知識加以分類、增補、重新歸類與重新脈絡化，就能帶出新的知識。現代的電腦系統就是一個鮮明的例子。這種「從顯性知識創造出顯性知識」的過程，被稱為「結合」。

The third and fourth modes of knowledge conversion relate to patterns of conversion involving both tacit and explicit knowledge. These conversion modes capture the idea that tacit and explicit knowledge are complementary and can expand over time through a process of mutual interaction. This interaction involves two different operations. One is **the conversion of tacit knowledge into explicit knowledge**, which will be called "externalization." The other is the conversion of explicit knowledge into tacit knowledge, which bears some similarity to the traditional notion of "learning" and will be referred to here as "internalization." As will be discussed later, "metaphor" plays an important role in the externalization process, and "action" is deeply related to the internalization process. Figure 1 illustrates the four modes of knowledge conversion.

► 中文譯文

第三種與第四種知識轉換的模式，跟「同時牽涉到隱性與顯性知識」的轉換型態有關。這兩種轉換模式捕捉到一個想法：隱性知識與顯性知識是互補的，而且能夠透過相互作用的過程，隨著時間擴展。這種相互作用牽涉到兩種不同的運作。一種是把隱性知識轉換成顯性知識，這會被稱為「外化」。另一種是把顯性知識轉換成隱性知識，它跟傳統「學習」的概念有些相似，在這裡會被稱為「內化」。如同後面會討論到的，「隱喻」在外化的過程裡扮演重要角色，而「行動」則跟內化的過程深切相關。圖 1 描繪了這四種知識轉換的模式。

Three of the four types of knowledge conversion—socialization, combination, and internalization, have partial analogs with aspects of organizational theory. For example, socialization is connected with theories of organizational culture, while combination is rooted in information processing and internalization has associations with organizational learning. By contrast, **the concept of externalization is not well developed**. The limited analysis that does exist is from the point of view of information creation (see Nonaka 1987).

► 中文譯文

在這四種知識轉換的型態裡，有三種，也就是社會化、結合與內化，跟組織理論的某些面向有著局部的對應。舉例來說，社會化跟組織文化的理論相連，結合則根植於資訊處理，而內化跟組織學習有所關聯。相對之下，「外化」這個概念則發展得並不充分。目前僅有的有限分析，是從「資訊創造」的角度切入的（見 Nonaka 1987）。

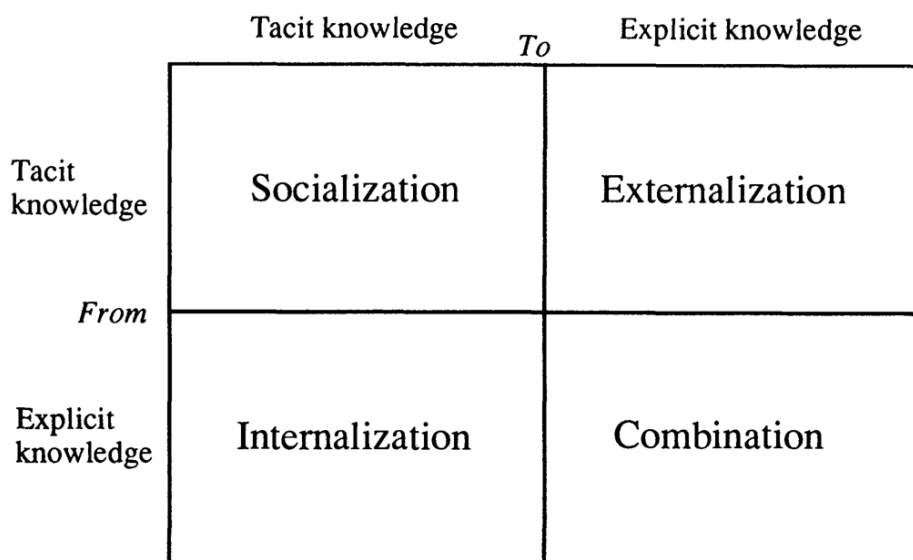


Figure 1. Modes of the Knowledge Creation

圖 1 知識創造的模式

Theories of organizational learning do not address the critical notion of externalization, and have paid little attention to the importance of socialization even though there has been an accumulation of research on "modeling" behavior in learning psychology. Another difficulty relates to the concepts of "double-loop learning" (Argyris and Schon 1978) or "unlearning" (Hedberg 1981), which arises from a strong orientation toward organization development (OD). Since the first integrated theory of organizational learning presented by Argyris and Schon, it has been widely assumed, implicitly or explicitly, that double-loop learning, i.e., the questioning and reconstruction of existing perspectives, interpretation frameworks, or decision premises, can be very difficult for organizations to implement by themselves. In order to overcome this difficulty, they argue that some kind of artificial intervention such as the use of organizational development programs is required. The limitation of this argument is that it assumes implicitly that someone inside or outside an organization knows "objectively" the right time and method for putting double-loop learning into practice. A mechanistic view of the organization lies behind this assumption. Seen from the vantage point of organizational knowledge creation, on the contrary, double-loop learning is not a special, difficult task but a daily activity for the organization. Organizations continuously create new knowledge by reconstructing existing perspectives, frameworks, or premises on a day-to-day basis. In other words, double-loop learning ability is "built into" the knowledge creating model, thereby circumventing the need to make unrealistic assumptions about the existence of a "right" answer.

► 中文譯文

組織學習的理論，並沒有處理「外化」這個關鍵概念，而且即使學習心理學裡已經累積了不少關於「示範」行為的研究，這些理論也很少關注社會化的重要性。另一個困難，跟「雙環學習」（Argyris and Schon 1978）或「反學習」（Hedberg 1981）這些概念有關，它們源自一種強烈傾向組織發展（OD）的取向。自從 Argyris 與 Schon 提出第一套整合性的組織學習理論以來，學界就普遍假定，不論明說或暗示：雙環學習，也就是對既有視角、詮釋框架或決策前提進行質疑與重建，對組織來說是很難靠自己落實的。為了克服這個困難，他們主張需要某種人為的介入，例如運用組織發展方案。這個論點的限制在於：它暗中假定，組織內部或外部有某個人，能夠「客觀地」知道把雙環學習付諸實行的正確時機與方法。這個假定的背後，是一種機械論式的組織觀。然而，從組織知識創造的角度來看，情況恰好相反：雙環學習並不是什麼特殊而困難的任務，而是組織每天都在做的日常活動。組織透過日復一日地重建既有的視角、框架或前提，持續不斷地創造新知識。換句話說，雙環學習的能力是「內建」在知識創造模型裡的，這樣一來，就不需要去做那種「假設存在一個『正確』答案」的不切實際假定了。

模式轉移與知識的螺旋 Modal Shift and Spiral of Knowledge

While each of the four modes of knowledge conversion can create new knowledge independently, the central theme of the model of organizational knowledge creation proposed here hinges on a dynamic interaction between the different modes of knowledge conversion. That is to say, knowledge creation centers on the building of both tacit and explicit

knowledge and, more importantly, on the interchange between these two aspects of knowledge through internalization and externalization.

► 中文譯文

雖然這四種知識轉換的模式，每一種都能獨立地創造新知識，但本文所提出的組織知識創造模型，其核心主軸卻在於「不同轉換模式之間的動態互動」。也就是說，知識創造的核心，在於同時建構隱性知識與顯性知識，而更重要的是，在於透過內化與外化，讓知識的這兩個面向彼此交流。

A failure to build a dialogue between tacit and explicit knowledge can cause problems. For example, both pure combination and socialization have demerits. A lack of commitment and neglect of the personal meaning of knowledge might mean that pure combination becomes a superficial interpretation of existing knowledge, which has little to do with here-and-now reality. It may also fail to crystallize or embody knowledge in a form that is concrete enough to facilitate further knowledge creation in a wider social context. The "sharability" of knowledge created by pure socialization may be limited and, as a result, difficult to apply in fields beyond the specific context in which it was created.

► 中文譯文

如果無法在隱性知識與顯性知識之間建立起對話，就可能引發問題。舉例來說，純粹的結合與純粹的社會化，都各有缺點。一旦缺乏承諾、又忽略了知識的個人意義，純粹的結合就可能淪為對既有知識的一種表面詮釋，跟「此時此地」的現實沒什麼關係。它也可能無法把知識結晶或具體化成一種「具體到足以促進更廣社會脈絡中進一步知識創造」的形式。而純粹社會化所創造出的知識，其「可分享性」可能有限，結果就是很難應用到「它被創造出來的那個特定脈絡」以外的領域。

Organizational knowledge creation, as distinct from individual knowledge creation, takes place when all four modes of knowledge creation are "organizationally" managed to form a continual cycle. This cycle is shaped by a series of shifts between different modes of knowledge conversion. There are various "triggers" that induce these shifts between different modes of knowledge conversion. First, the socialization mode usually starts with the building of a "team" or "field" of interaction. This field facilitates the sharing of members' experiences and perspectives. Second, the externalization mode is triggered by successive rounds of meaningful "dialogue." In this dialogue, the sophisticated use of "metaphors" can be used to enable team members to articulate their own perspectives, and thereby reveal hidden tacit knowledge that is otherwise hard to communicate. Concepts formed by teams can be combined with existing data and external knowledge in a search of more concrete and sharable specifications. This combination mode is facilitated by such triggers as "coordination" between team members and other sections of the organization and the "documentation" of existing knowledge. Through an iterative process of trial and error, concepts are articulated and developed until they emerge in a concrete form. This "experimentation" can trigger internalization through a process of "learning by doing." Participants in a "field" of action share explicit knowledge that is gradually translated, through interaction and a process of trial-and-error, into different aspects of tacit knowledge.

► 中文譯文

組織知識創造跟個人知識創造不同，它發生的時機，是當這四種知識創造模式都被「以組織的方式」管理，進而形成一個持續不斷的循環時。這個循環，是由不同知識轉換模式之間一連串的轉移所形塑出來的。有各式各樣的「觸發器」，會誘發這些不同模式之間的轉移。第一，社會化模式通常是從建立一個互動的「團隊」或「場域」開始的。這個場域促進了成員之間經驗與視角的分享。第二，外化模式是被一輪又一輪有意義的「對話」所觸發的。在這種對話裡，巧妙地運用「隱喻」，可以讓團隊成員把自己的視角清楚表述出來，藉此把那些原本難以溝通、隱藏著的隱性知識揭露出來。團隊所形成的概念，可以跟既有的資料及外部知識相結合，去尋找更具體、更可分享的規格。這種結合模式，是被像「團隊成員與組織其他部門之間的協調」、以及「既有知識的文件化」這類觸發器所促進的。透過反覆試誤的循環過程，概念被清楚表述、不斷發展，直到它們以具體的形式浮現出來。這種「實驗」，能夠透過一個「做中學」的過程，去觸發內化。一個行動「場域」裡的參與者，會分享顯性知識，而這些顯性知識會透過互動與試誤的過程，逐漸被轉譯成隱性知識的不同面向。

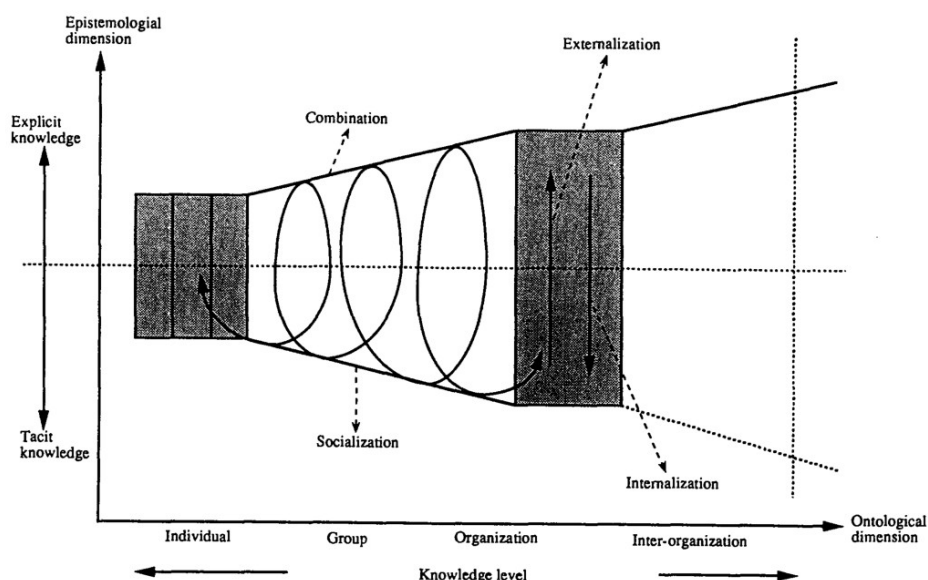


Figure 2. Spiral of Organizational Knowledge Creation

圖 2 組織知識創造的螺旋

While tacit knowledge held by individuals may lie at the heart of the knowledge creating process, realizing the practical benefits of that knowledge centers on its externalization and amplification through dynamic interactions between all four modes of knowledge conversion. **Tacit knowledge is thus mobilized** through a dynamic "entangling" of the different modes of knowledge conversion in a process which will be referred to as a "spiral" model of knowledge creation, illustrated in Figure 2. The interactions between tacit knowledge and explicit knowledge will tend to become larger in scale and faster in speed as more actors in and around the organization become involved. Thus, organizational knowledge creation can be viewed as an upward spiral process, starting at the individual level moving up to the collective (group) level, and then to the organizational level, sometimes reaching out to the interorganizational level.

中文譯文

雖然個人手上握有的隱性知識，可能位於知識創造過程的核心，但要實現這種知識的實際效益，關鍵卻在於透過四種轉換模式之間的動態互動，去把它「外化」並「擴大」。隱性知識就是這樣被動員起來的：透過不同知識轉換模式之間動態的「交纏」，在一個會被稱為知識創造「螺旋」模型的過程裡（如圖 2 所示）被調動。當組織內部及周邊有越來越多的行動者參與進來，隱性知識與顯性知識之間的互動，規模會傾向變得更大，速度也會變得更快。因此，組織知識創造可以看成一個向上的螺旋過程：它從個人層次起步，往上到集體（群體）層次，再到組織層次，有時還會延伸到組織之間的層次。

2.5 從隱喻到模型：知識創造的方法論 2.5 From Metaphor to Model: Methodology of Knowledge Creation

Before concluding this presentation of the basic constructs of the theory, it is helpful to consider some general principles for facilitating the management of knowledge conversion. One effective method of converting tacit knowledge into explicit knowledge is the use of metaphor. As Nisbet (1969, p. 5) noted, "(m)uch of what Michael Polanyi has called 'tacit knowledge' is expressible—in so far as it is expressible at all—in metaphor." "The essence of metaphor is understanding and experiencing one kind of thing in terms of another (Lakoff and Johnson 1980, p. 5)." Even though the metaphor is not in itself a thinking process, it enables us to experience a new behavior by making inferences from the model of another behavior. The use of metaphor is broader than the traditional, lexical definition of the term (meta = change; phor = move). According to Lakoff and Johnson: "metaphor is pervasive in everyday life, not just in language but in thought and action. Our ordinary conceptual system, in terms of which we both think and act, is fundamentally metaphorical in nature" (Lakoff and Johnson 1980, p. 3).

中文譯文

在結束這套理論基本構念的說明之前，先思考一些「促進知識轉換管理」的一般性原則，會很有

幫助。把隱性知識轉換成顯性知識的一個有效方法，就是運用隱喻。如同 Nisbet (1969, p. 5) 所指出的：「Michael Polanyi 所說的『隱性知識』，其中有很大一部分是可以被表達出來的，就它終究能被表達的程度而言，是透過隱喻來表達的。」「隱喻的本質，是用某一種事物去理解並體驗另一種事物 (Lakoff and Johnson 1980, p. 5)。」即使隱喻本身並不是一種思考過程，它仍然讓我們能夠透過「從另一種行為的模型進行推論」，去體驗一種新的行為。隱喻的運用，比這個詞傳統的字面定義 (meta=改變; phor=移動) 要來得更廣。根據 Lakoff 與 Johnson 的說法：「隱喻在日常生活中無所不在，不只存在於語言中，也存在於思考與行動裡。我們用來思考與行動的那套日常概念系統，本質上根本就是隱喻性的」 (Lakoff and Johnson 1980, p. 3)。

As a method of perception, metaphor depends on imagination and intuitive learning through symbols, rather than on the analysis or synthesis of common attributes shared by associated things. Rosch (1973) suggested that man describes the world, not in the formal attributes of concepts, but in terms of prototypes. For example, the robin could be seen as a better prototype than the turkey for a small bird. Prototypes provide a mechanism for recognizing the maximum level of information with a minimum of energy.

► 中文譯文

作為一種感知的方法，隱喻倚賴的是想像力、以及透過符號進行的直覺式學習，而不是去分析或綜合「相關事物所共有的共同屬性」。Rosch (1973) 提出，人描述世界的方式，不是用概念的形式屬性，而是用「原型」。舉例來說，就「小型鳥」這個概念而言，知更鳥會比火雞更適合當作原型。原型提供了一套機制，讓我們能用最少的能量，去辨認出大量的資訊。

Metaphor is not merely the first step in transforming tacit knowledge into explicit knowledge; it constitutes an important method of creating a network of concepts which can help to generate knowledge about the future by using existing knowledge. Metaphor may be defined as being 'two contradicting concepts incorporated in one word.' It is a creative, cognitive process which relates concepts that are far apart in an individual's memory. While perception through prototype is in many cases limited to concrete, mundane concepts, metaphor plays an important role in associating abstract, imaginary concepts. When two concepts are presented in a metaphor, it is possible not only to think of their similarity, but also to make comparisons that discern the degree of imbalance, contradiction or inconsistency involved in their association. The latter process becomes the basis for creating new meaning.

► 中文譯文

隱喻不只是「把隱性知識轉化成顯性知識」的第一步而已，它本身就構成一種重要的方法，用來創造一個概念的網絡，而這個網絡能幫助我們運用既有知識去產生關於未來的知識。隱喻可以被定義成「把兩個相互矛盾的概念融合進同一個詞裡」。它是一種具創造性的認知過程，把個人記憶中相隔很遠的概念連結起來。透過原型來感知，在許多情況下侷限於具體的、日常的概念；而隱喻則在「連結抽象的、想像性的概念」上扮演重要角色。當兩個概念被放進一個隱喻裡呈現時，我們不只能去思考它們的相似之處，還能進行比較，去辨別它們的連結之中所牽涉到的失衡、矛盾或不一致的程度。而後面這個過程，正是創造新意義的基礎。

According to Bateson (1973) metaphors cut across different contexts and thus allow imaginative perceptions to combine with literal levels of cognitive activities. This experience, he further argues, will promote the type of "presupposition-negation" learning that is closely related with the formation of new paradigms.

► 中文譯文

根據 Bateson (1973) 的說法，隱喻會橫切過不同的脈絡，因而讓富想像力的感知，能夠跟認知活動中字面層次的部分結合起來。他進一步主張，這種經驗會促進一種「預設否定」式的學習，而這種學習跟新典範的形成密切相關。

Contradictions incorporated in metaphor may be harmonized through the use of analogies. Analogy reduces ambiguity by highlighting the commonness of two different things. Metaphor and analogy are often confused. The association of meanings by metaphor is mostly driven by intuition, and involves images. On the other hand, the association of meanings

through analogy is more structural/functional and is carried out through rational thinking. As such, metaphors provide much room for free association (discontinuity). Analogy allows the functional operation of new concepts or systems to be explored by reference to things that are already understood. In this sense, an analogy—that enables us to know the future through the present—assumes an intermediate role in bridging the gap between image and logic.

► 中文譯文

隱喻裡所融合的種種矛盾，可以透過運用「類比」來加以調和。類比藉由凸顯兩個不同事物之間的共通之處，來降低模糊性。隱喻與類比經常被混為一談。透過隱喻來連結意義，大多是由直覺所驅動的，而且牽涉到意象。另一方面，透過類比來連結意義，則比較偏結構性與功能性，是透過理性思考來進行的。正因如此，隱喻提供了很大的自由聯想（不連續）空間。而類比，則讓我們能夠藉由參照「已經被理解的事物」，去探索新概念或新系統的功能性運作。從這個意義上說，類比，這個讓我們得以「透過現在去認識未來」的東西，扮演了一個居間的角色，去搭起意象與邏輯之間的橋樑。

It follows from the preceding discussion that tacit knowledge may be transformed into explicit knowledge by (1) recognizing contradictions through metaphor, and (2) resolving them through analogy. Explicit knowledge represents a model within which contradictions are resolved and concepts become transferable through consistent and systematic logic. In the business organization, a typical model is the prototype that represents the product concept. The prototype's specification is then explicit knowledge. It has been pointed out that metaphor, analogy, and model are all part of the process of scientific discovery. Whether the metaphor-analogy-model sequence is indispensable in all such processes will depend upon the nature of the question under study; yet in creating new concepts, the model is usually generated from a metaphor.

► 中文譯文

從前面的討論可以得出：隱性知識之所以能被轉化成顯性知識，是透過（1）藉由隱喻來辨認出矛盾，以及（2）藉由類比來化解這些矛盾。顯性知識代表的是一個模型，在這個模型裡，矛盾被化解，而概念也透過一致而有系統的邏輯，變得可以被傳遞。在企業組織裡，一個典型的模型，就是代表產品概念的「原型」。而原型的規格，就是顯性知識。已經有人指出，隱喻、類比與模型，全都是科學發現過程的一部分。至於「隱喻到類比再到模型」這個順序，是不是在所有這類過程中都不可或缺，則要看所研究問題的性質而定；不過，在創造新概念時，模型通常是從隱喻裡產生出來的。

3. 組織知識創造的過程 3. The Process of Organizational Knowledge Creation

The theoretical constructs and models described in §2 may now be related to organizational knowledge creation in a corporate organizational setting. This will be approached by assessing the processes that enable individual knowledge to be enlarged, amplified, and justified within an organization.

► 中文譯文

第 2 節所描述的理論架構與模型，現在可以連結到企業組織情境裡的組織知識創造。具體的做法，是去檢視那些讓個人知識能在組織裡被擴大、被放大、被證成的過程。

3.1 個人知識的擴大 3.1 The Enlargement of an Individual's Knowledge

The prime mover in the process of organizational knowledge creation is the individual. Individuals accumulate tacit knowledge through direct "hands-on" experience. The quality of that tacit knowledge is influenced by two important factors. One factor is the "variety" of an individual's experience. If this experience is limited to routine operations, the amount of tacit knowledge obtained from monotonous and repetitive tasks will tend to decrease over time. Routine tasks mitigate against creative thinking and the formation of new knowledge. However, increasing the variety of experience is not sufficient by itself to raise the quality of tacit knowledge. If the individual finds various experiences to be completely

unrelated, there will be little chance that they can be integrated to create a new perspective. What matters is "high quality" experience which might, on occasion, involve the complete redefinition of the nature of a "job."

► 中文譯文

組織知識創造過程裡的原動力，是個人。個人透過親自動手的第一手經驗，累積隱性知識。而這些隱性知識的品質，受兩個重要因素影響。第一個因素，是個人經驗的「多樣性」。如果經驗只侷限在例行操作，那麼從單調、重複的工作裡得到的隱性知識，會隨時間慢慢變少。例行工作會壓抑創造性思考，也不利於新知識的形成。但是，光是增加經驗的多樣性，還不足以提升隱性知識的品質。如果個人覺得這些不同的經驗彼此完全無關，那它們幾乎沒機會被整合起來，去創造出一個新的觀點。真正關鍵的，是「高品質」的經驗，這種經驗有時甚至會徹底重新定義一份「工作」的本質。

A second factor that determines the quality of tacit knowledge is "knowledge of experience." The essence of "knowledge of experience" is an embodiment of knowledge through a deep personal commitment into bodily experience. Varela et al. (1991) have pointed out that the embodied nature of human knowledge has long been neglected in Western epistemological traditions that have followed from Descartes. They define embodiment as: "a reflection in which body and mind have been brought together" (1991, p. 27). Yuasa (1987) describes this "oneness of body-mind" as the free state of minimal distance between movement of the mind and of the body, as for example in the dynamic performance of a master actor on a stage (1987, p. 28). As Merleau-Ponty (1964) pointed out, bodily experience plays a critical role in the process of crystallization. Commitment to bodily experience means an intentional self-involvement in the object and situation which transcends the subject-object distinction, thereby providing access to "pure experience" (Nishida 1960). This notion is prevalent in oriental culture. As Yuasa mentions:

► 中文譯文

決定隱性知識品質的第二個因素，是「經驗的知識」。「經驗的知識」的核心，是透過深刻的個人投入，把知識體現在身體經驗裡。Varela 等人（1991）指出，人類知識這種「身體化」的本質，長久以來在承襲自 Descartes（笛卡兒）的西方知識論傳統裡一直被忽略。他們把「身體化」定義為：「一種把身與心重新合在一起的反思」（1991, p.27）。Yuasa（湯淺，1987）把這種「身心合一」描述為一種自在的狀態，也就是心的活動與身的活動之間距離最小的狀態，例如一位大師級演員在舞台上充滿張力的演出（1987, p.28）。如同 Merleau-Ponty（梅洛龐蒂，1964）所指出的，身體經驗在「結晶化」的過程裡扮演關鍵角色。對身體經驗的投入，意味著刻意地把自己投進對象與情境裡，超越了主體與客體的分別，因而能通往「純粹經驗」（Nishida 西田，1960）。這個觀念在東方文化裡相當普遍。如同 Yuasa 所說：

One revealing characteristic of the philosophical uniqueness of Eastern thought is presupposed in the philosophical foundation of the Eastern theories. To put it simply, true knowledge cannot be obtained simply by means of theoretical thinking, but only through "bodily recognition or realization" (tainin or taitoku), that is, through the utilization of one's total mind and body. Simply stated, this is to "learn with the body" not the brain. Cultivation is a practice that attempts, so to speak, to achieve true knowledge by means of one's total mind and body (1987, pp. 25-26).

► 中文譯文

東方思想在哲學上的獨特性，有一個很能說明問題的特徵，預設在東方理論的哲學基礎裡。簡單說，真正的知識沒辦法只靠理論性的思考得到，而只能透過「身體的認識或體悟」（日文：体認 tainin、或體得 taitoku），也就是透過運用一個人完整的身與心去取得。講白一點，這是用「身體」去學，而不是用腦袋去學。所謂的修養，可以說就是一種試圖以一個人完整的身心去達到真知的實踐（1987, pp.25-26）。

A good case in point is "on-the-spot-ism" in Japanese management. In developing the products and identifying the markets, Japanese firms encourage the use of judgement and knowledge formed through interaction with customers—and by personal bodily experience rather than by "objective," scientific conceptualization. Social interaction between individuals, groups and organizations are fundamental to organizational knowledge creation in Japan. Nevertheless,

since this approach uses hands-on experience and action, it sometimes falls in the category of "experientism" which neglects the importance of reflection and logical thinking. It tends to overemphasize action and efficiency at the expense of a search for higher level concepts which have universal application.

► 中文譯文

一個很好的例子，是日本管理裡的「現場主義」（on-the-spot-ism）。在開發產品、辨識市場時，日本企業鼓勵運用那種透過跟顧客互動、以及透過個人親身的身體經驗所形成的判斷與知識，而不是靠「客觀的」、科學式的概念化。在日本，個人、群體與組織之間的社會互動，是組織知識創造的根本。不過，因為這種做法很依賴親身的經驗與行動，它有時會落入「經驗主義」的範疇，忽略了反思與邏輯思考的重要性。它往往過度強調行動與效率，卻犧牲了對那些具有普遍適用性的更高層次概念的追尋。

While the concepts of "high-quality experience" and "knowledge of experience" may be used to raise the quality of tacit knowledge, they have to be counterbalanced by a further approach to knowledge creation that raises the quality of explicit knowledge. Such an approach may be called a "knowledge of rationality," which describes a rational ability to reflect on experience. Knowledge of rationality is an explicit-knowledge-oriented approach that is dominant in Western culture. It centers on the "combination" mode of knowledge conversion, and is effective in creating digital, discrete declarative knowledge. Knowledge of rationality tends to ignore the importance of commitment, and instead centers a reinterpretation of existing explicit knowledge.

► 中文譯文

「高品質經驗」與「經驗的知識」這兩個概念雖然可以用來提升隱性知識的品質，但它們必須由另一種「提升顯性知識品質」的知識創造取向來加以平衡。這種取向可以稱為「理性的知識」，指的是一種反思經驗的理性能力。理性的知識是一種以顯性知識為導向的取向，在西方文化裡居於主導地位。它以知識轉換的「combination（組合）」模式為核心，擅長創造出數位的、離散的、陳述性的知識。理性的知識往往忽略「投入」的重要性，反而把重心放在對既有顯性知識的重新詮釋上。

In order to raise the total quality of an individual's knowledge, the enhancement of tacit knowledge has to be subjected to a continual interplay with the evolution of relevant aspects of explicit knowledge. In this connection, Schon (1983) pointed out the importance of "reflection in action," i.e., reflecting while experiencing. Individual knowledge is enlarged through this interaction between experience and rationality, and crystallized into a unique perspective original to an individual. These original perspectives are based on individual belief and value systems, and will be a source of varied interpretations of shared experience with others in the next stage of conceptualization.

► 中文譯文

為了提升一個人知識的整體品質，隱性知識的增強必須跟相關顯性知識的演進，維持一種持續的相互作用。在這一點上，Schon（1983）指出「行動中反思」（reflection in action）的重要性，也就是一邊經驗、一邊反思。個人知識正是透過這種經驗與理性之間的互動而被擴大，並結晶成一個對該個人來說獨一無二的、原創的觀點。這些原創觀點奠基於個人的信念與價值體系，並且會在下一個階段，也就是概念化的階段裡，成為對「與他人共享的經驗」做出各種不同詮釋的來源。

3.2 分享隱性知識與概念化 3.2 Sharing Tacit Knowledge and Conceptualization

As we saw in the previous section, the process of organizational knowledge creation is initiated by the enlargement of an individual's knowledge within an organization. The interaction between knowledge of experience and rationality enables individuals to build their own perspectives on the world. Yet these perspectives remain personal unless they are articulated and amplified through social interaction. One way to implement the management of organizational knowledge creation is to create a "field" or "self-organizing team" in which individual members collaborate to create a new concept.

► 中文譯文

如同我們在上一節看到的，組織知識創造的過程，是由組織內個人知識的擴大所啟動的。「經驗的知識」與「理性的知識」之間的互動，讓個人能建立起自己對世界的觀點。然而，除非這些觀點透過社會互動被表達出來、被放大，否則它們就只會停留在個人層次。要落實對組織知識創造的管理，方法之一，是創造一個「場域（field）」或「自組織團隊（self-organizing team）」，讓成員們在裡面協力創造出新概念。

In this connection, it is helpful to draw on the concept of an organization's "mental outlook" as articulated in Sandelands and Stablein's (1987) pioneering work on "organizational mind." While making caveats about the dangers of reification and anthropomorphism, these authors use the analogy of "mind" to identify the process by which organizations form ideas. Mind is distinct from the brain in the same way that computer software is distinct from hardware. Against this background, intelligence may be seen as the ability to maintain a working similarity between mind and nature.

► 中文譯文

在這一點上，借用 Sandelands 與 Stablein（1987）在「組織心智（organizational mind）」這項開創性研究裡所闡述的「組織心智樣貌」概念，會很有幫助。這兩位作者一邊提醒大家要小心「具體化」與「擬人化」的風險，一邊用「心智」這個類比，去指認組織形成想法的過程。心智之於大腦，就好比電腦軟體之於硬體，兩者是有區別的。在這個背景下，智能可以被看成是一種能力，也就是讓心智與自然之間維持某種「運作上的相似」的能力。

The development of ideas associated with organizational mind requires some form of physical substrate (i.e. hardware) which Sandelands and Stablein (1987) argue might be derived from "patterns of behavior traced by people and machines" (p. 139). Organizational behaviors can convey ideas and, like the firing of neurons in the brain, may trigger other behaviors and so form a trace of activation.

► 中文譯文

跟組織心智有關的想法要發展出來，需要某種形式的物理基質（也就是硬體），而 Sandelands 與 Stablein（1987）認為，這種基質可能來自「由人與機器所描繪出來的行為模式」（p.139）。組織行為能夠傳遞想法，而且就像大腦裡神經元的放電一樣，可能會觸發其他行為，於是形成一條「活化的軌跡」。

In the brain, whether or not one neuron influences another depends on a complex set of factors having primarily to do with physical proximity, availability of pathways, intensity of the electrochemical signal, and whether or not the target neuron is inhibited by other neurons. Similarly, whether one behavior influences another in social organizations depends on a complex of factors primarily concerned with physical access, lines of communication, power, and competition from other behaviors. At an abstract formal level, at least, the politics of the social organization and the physiology of the brain share much in common (Sandelands and Stablein 1987, p. 140).

► 中文譯文

在大腦裡，一個神經元會不會影響另一個神經元，取決於一組複雜的因素，主要跟物理上的鄰近程度、傳導路徑是否暢通、電化學訊號的強度，以及目標神經元有沒有被其他神經元抑制有關。同樣地，在社會組織裡，一個行為會不會影響另一個行為，也取決於一組複雜的因素，主要跟物理上能否接觸、溝通管道、權力，以及來自其他行為的競爭有關。至少在抽象的形式層次上，社會組織的政治運作，跟大腦的生理運作，有很多共通之處（Sandelands and Stablein 1987, p.140）。

It is human activity that creates organizational mind as individuals interact and trigger behavior patterns in others. Managing a self-organizing team involves building an appropriate degree of flexibility into the system which can accommodate a diversity of imaginative thinking in the pursuit of new problems and solutions.

► 中文譯文

真正創造出組織心智的，是人的活動，因為個人會彼此互動，並在他人身上觸發行為模式。管理一個自組織團隊，意味著要在系統裡建立適當程度的彈性，好讓系統能容納各式各樣富有想像力的思考，去追尋新的問題與新的解法。

建構場域：打造自組織團隊 Constructing a Field: Building a Self-organizing Team

To bring personal knowledge into a social context within which it can be amplified, it is necessary to have a "field" that provides a place in which individual perspectives are articulated, and conflicts are resolved in the formation of higher-level concepts. Berger and Luchman (1966) say that reality in everyday life is socially constructed. Individual behavior ought to be relativized through an interactive process to construct "social reality."

► 中文譯文

要把個人知識帶進一個能讓它被放大的社會脈絡裡，就必須要有一個「場域」，提供一個地方，讓個人的觀點能被表達出來，也讓衝突能在形成更高層次概念的過程中被化解。Berger 與 Luchman (1966) 說，日常生活裡的真實，是被社會建構出來的。個人行為應該透過一個互動的過程被「相對化」，以建構出「社會真實」。

In the business organization, the field for interaction is often provided in the form of an autonomous, self-organizing "team" made of several members coming from a variety of functional departments. It is a critical matter for an organization to decide when and how to establish such a "field" of interaction in which individuals can meet and interact. It defines "true" members of knowledge creation and thus clarifies the domain in which perspectives are interacted.

► 中文譯文

在企業組織裡，互動的場域常常以一個自主、自組織的「團隊」形式出現，這個團隊由來自不同功能部門的數名成員組成。對組織來說，要決定「何時」以及「如何」建立這樣一個讓個人能碰面、互動的「場域」，是一件關鍵的事。因為它界定了知識創造的「真正」成員是誰，也因此釐清了觀點在哪個範圍裡彼此交流。

The team needs to be established with regard to the principles of self-organization. In Lewin's (1951) development of the field theory in social psychology, a group is defined as "a dynamic whole based on interdependence rather on similarity." Some indication of the number of members and the composition of their background can be achieved using the principle of "requisite variety" (Ashby 1956). According to our observation of successful project teams in Japanese firms, the appropriate team size may be in the region between 10 and 30 individuals, with an upper limit arising because direct interaction between all the group members tends to decrease as group size increases. Within the team, there are usually 4 to 5 "core" members who have career histories that include multiple job functions. These core members form focal points in the team and could be seen as the organizational equivalent of the central element in a series of nested Russian dolls. That is to say there is a radial pattern of interaction with other members, with closer links being associated with key individuals. Core members play a critical role in assuring appropriate "redundancy" of information within the cross-functional team. Other attributes of members such as formal position, age, gender, etc. might be determined with regard to Morgan's (1986) four principles of "learning to learn, requisite variety, minimum critical specification, and redundancy of functions."

► 中文譯文

團隊的建立，必須顧及自組織的原則。在 Lewin (1951) 對社會心理學「場域理論」的發展裡，一個群體被定義為「一個建立在『相互依賴』而非『相似性』之上的動態整體」。至於成員的人數，以及他們背景該如何組成搭配，可以用「必要多樣性 (requisite variety)」這個原則 (Ashby 1956) 抓出一些方向。根據我們對日本企業裡成功專案團隊的觀察，適當的團隊規模大約落在 10 到 30 人之間，之所以有人數上限，是因為當群體規模變大，所有成員之間的直接互動往往會跟著減少。在團隊裡，通常會有 4 到 5 位「核心」成員，他們的職涯經歷橫跨多種職務功能。這些核心成員在團隊裡形成一個個焦點，可以把他們看成是組織版的「俄羅斯娃娃」裡，那一層層套疊結構最中央的核心。也就是說，成員之間的互動呈現一種放射狀的模式，跟關鍵人

物的連結會更緊密。核心成員在確保跨功能團隊裡資訊有適當的「冗餘 (redundancy)」這件事上，扮演關鍵角色。至於成員其他的屬性，例如正式職位、年齡、性別等等，則可以參照 Morgan (1986) 的四個原則來決定，也就是「學習如何學習、必要多樣性、最低限度的關鍵規格，以及功能的冗餘」。

The span of team activities need not confined to the narrow boundary of the organization. Rather, it is a process that frequently makes extensive use of knowledge in environment, especially that of customers and suppliers. As Norman (1988) argues, the mental outlook of an organization is shaped by a complex pattern of factors within and outside the organization. In some Japanese firms, for example, suppliers of parts and components are sometimes involved in the early stages of the product development. The relationship between manufacturers and suppliers is less hierarchial and arms length than in Western countries. Some other Japanese companies involve customers in the field of new product planning. In both cases, sharing tacit knowledge with suppliers or customers through coexperience and creative dialogue play a critical role in creating relevant knowledge.

► 中文譯文

團隊活動的範圍，不必侷限在組織狹窄的邊界之內。相反地，這是一個經常會大量運用環境中知識的過程，尤其是顧客與供應商的知識。如同 Norman (1988) 所主張的，一個組織的心智樣貌，是由組織內部與外部一組複雜的因素交織塑造出來的。舉例來說，在某些日本企業裡，零件與組件的供應商有時會被拉進產品開發的早期階段。製造商與供應商之間的關係，比起西方國家，沒那麼層級分明，也沒那麼保持距離。另外有些日本公司，則會讓顧客參與新產品規劃的場域。在這兩種情況下，透過「共經驗 (coexperience)」與創造性對話，跟供應商或顧客分享隱性知識，在創造出相關知識這件事上都扮演關鍵角色。

The significance of links between individuals that span boundaries, both within and outside the organization, has been highlighted by Brown and Duguid's (1991) revealing insight into the operation of "evolving communities of practice." These communities reflect the way in which people actually work as opposed to the formal job descriptions or task-related procedures that are specified by the organization. Attempts to solve practical problems often generate links between individuals who can provide useful information. The exchange and development of information within these evolving communities facilitate knowledge creation by linking the routine dimensions of day-to-day work to active learning and innovation. Collaboration to exchange ideas through shared narratives and "war stories" can provide an important platform on which to construct shared understanding out of conflicting and confused data.

► 中文譯文

個人之間那些跨越邊界（包括組織內部與外部）的連結有多重要，Brown 與 Duguid (1991) 對「演化中的實踐社群 (evolving communities of practice)」運作方式的精闢洞見，把這一點凸顯了出來。這些社群反映的是人們「實際上」怎麼工作，而不是組織所規定的正式職務說明、或跟任務有關的標準程序。人們試圖解決實務問題時，常常會在那些能提供有用資訊的個人之間，自然產生連結。在這些演化中的社群裡，資訊的交流與發展，把日常工作裡的例行面向，連結到主動的學習與創新，從而促進了知識創造。透過共享的敘事、以及「戰場故事 (war stories)」來交換想法的這種協作，能提供一個重要的平台，讓人們可以從相互矛盾、混亂的資料裡，建構出共享的理解。

By contrast with conceptions of groups as bounded entities within an organization, evolving communities of practice are "more fluid and interpenetrative than bounded, often crossing the restrictive boundaries of the organization to incorporate people from outside" (Brown and Duguid 1991, p. 49). Moreover, these communities can provide important contributions to visions for future development. Thus these communities represent a key dimension to socialization and its input to the overall knowledge creation process.

► 中文譯文

相對於那種把群體看成組織內部「有明確邊界的實體」的看法，演化中的實踐社群「比起『有邊界』更顯得流動、相互滲透，而且常常跨越組織那些限制性的邊界，把外部的人也納進來」

(Brown and Duguid 1991, p.49)。不只如此，這些社群還能對「未來發展的願景」做出重要貢獻。因此，這些社群代表了「socialization（社會化）」的一個關鍵面向，以及它對整體知識創造過程的投入。

The self-organizing team triggers organizational knowledge creation through two processes. First, it facilitates the building of mutual trust among members, and accelerates creation of an implicit perspective shared by members as tacit knowledge. The key factor for this process is sharing experience among members. Second, the shared implicit perspective is conceptualized through continuous dialogue among members. This creative dialogue is realized only when redundancy of information exists within the team. The two processes appear simultaneously or alternatively in the actual process of knowledge creation within a team.

► 中文譯文

自組織團隊透過兩個過程，觸發組織知識創造。第一，它促成成員之間建立起相互信任，並加速催生出一種由成員以隱性知識形式共享的「內隱觀點」。這個過程的關鍵因素，是成員之間「分享經驗」。第二，這個共享的內隱觀點，會透過成員之間持續的對話被「概念化」。而這種創造性的對話，唯有在團隊內部存在資訊冗餘時，才有可能實現。在團隊知識創造的實際過程裡，這兩個過程會同時出現，或者交替出現。

Before discussing these two processes further, it is necessary to mention another dimension of the knowledge creating process that can be associated with the self-organizing team. Schefflen (1982) proposed an idea of "interaction rhythms," in which social interactions were viewed as being both simultaneous and sequential. The management of interaction rhythms among team members, i.e., that of divergence and convergence of various interaction rhythms, plays a critical role in accelerating the knowledge creation process. Within the team, rhythms of different speed are first generated and amplified up to certain point of time and level, and then are given momentum for convergence towards a concept. Therefore, the crucial role of the team leader concerns how to balance the rhythm of divergence and convergence in the process of dialogues and shared experience.

► 中文譯文

在進一步討論這兩個過程之前，有必要先提一下知識創造過程的另一個面向，它同樣跟自組織團隊有關。Schefflen (1982) 提出了「互動節奏 (interaction rhythms)」的概念，把社會互動同時看成是「同時發生的」與「依序進行的」。管理團隊成員之間的互動節奏，也就是管理各種互動節奏的「發散」與「收斂」，在加速知識創造過程上扮演關鍵角色。在團隊裡，不同速度的節奏會先被產生、被放大，一直到某個時間點與某個程度，然後再被賦予動能，朝著一個概念去收斂。因此，團隊領導者的關鍵角色，就在於如何在對話與分享經驗的過程裡，去平衡發散與收斂的節奏。

In sum, the cross-functional team in which experience sharing and continuous dialogue are facilitated by the management of interaction rhythms serves as the basic building block for structuring the organization knowledge creation process. The team is different from a mere group in that it induces self-organizing process of the entire organization through which the knowledge at the group level is elevated to the organizational level.

► 中文譯文

總的來說，一個透過互動節奏的管理來促成「經驗分享」與「持續對話」的跨功能團隊，就是建構組織知識創造過程的基本單元。這種團隊跟單純的「群體」不一樣，因為它會引發整個組織的自組織過程，並透過這個過程，把群體層次的知識，提升到組織層次。

分享經驗 Sharing Experience

In order for the self-organizing team to start the process of concept creation, it first needs to build mutual trust among members. As we shall see later, **concept creation involves a difficult process of externalization**, i.e., converting tacit knowledge **(which by nature is hard to articulate)** into an explicit concept. This challenging task involves repeated, time-consuming dialogue among members. Mutual trust is an indispensable base for facilitating this type of constructive "collaboration" (Schrage 1990). A key way to build mutual trust is to share one's original experience—the fundamental

source of tacit knowledge. Direct understanding of other individuals relies on shared experience that enables team members to "indwell" into others and to grasp their world from "inside."

► 中文譯文

自組織團隊要開始「概念創造」的過程，首先得在成員之間建立相互信任。我們稍後會看到，概念創造牽涉到一個困難的「externalization（外化）」過程，也就是把隱性知識（這種知識本質上就很難講清楚）轉換成一個顯性的概念。這項充滿挑戰的任務，需要成員之間一再進行、相當耗時的對話。而相互信任，正是促成這種建設性「協作」不可或缺的基礎（Schrage 1990）。建立相互信任的一個關鍵做法，是分享一個人最原初的經驗，那是隱性知識最根本的來源。要直接理解另一個人，靠的是共享的經驗，因為它能让團隊成員「寓居（indwell）」進別人裡面，從「內部」去把握對方的世界。

Shared experience also facilitates the creation of "common perspectives" which can be shared by team members as a part of their respective bodies of tacit knowledge. The dominant mode of knowledge conversion involved here is socialization. Various forms of tacit knowledge that are brought into the field by individual members are converted through coexperience among them to form a common base for understanding.

► 中文譯文

共享的經驗也有助於創造出「共同觀點」，讓團隊成員能把它當成各自隱性知識體系的一部分來共享。這裡牽涉到的主要知識轉換模式，是「socialization（社會化）」。各位成員帶進場域裡的各種形式的隱性知識，會透過他們之間的「共經驗」被轉換，從而形成一個共同的理解基礎。

As was mentioned earlier, tacit knowledge is a distinctly personal concept. Varela et al. (1991) point out the limitation of the cognitivist view of human experience in comparison with the non-Western philosophical view, and suggest that cognitive experience is "embodied action" rather than a mere representation of a world that exists independent of our cognitive system. The mutual conversion of such embodied, tacit knowledge is accelerated by synchronizing both body and mind in the face of the same experience. Coexperience with others enables us to transcend the ordinary "I-Thou" distinction, and opens up the world of common understanding, which Schefflen (1982) called "Field Epistemology." Condon (1976) shared this view that communication is a simultaneous and contextual phenomenon in which people feel a change occurring, share the same sense of change, and are moved to take action. In other words, communication is like a wave that passes through people's bodies and culminates when everyone synchronizes himself with the wave. Thus, the sharing of mental and physical rhythm among participants of a field may serve as the driving force of socialization.

► 中文譯文

如同前面提過的，隱性知識是一個極為個人化的概念。Varela 等人（1991）指出，比起非西方的哲學觀點，認知主義對人類經驗的看法是有侷限的，他們主張認知經驗是一種「體現的行動（embodied action）」，而不只是對一個「獨立於我們認知系統之外而存在的世界」的單純再現。這種體現的、隱性的知識要彼此轉換，會因為在面對同一個經驗時「身與心同步」而加速。跟他人共經驗，讓我們能超越平常那種「我與你（I-Thou）」的分別，打開一個共同理解的世界，Schefflen（1982）把這個世界稱為「場域知識論（Field Epistemology）」。Condon（1976）也認同這個看法，認為溝通是一種同時發生的、依脈絡而定的現象，在這個現象裡，人們感覺到某種變化正在發生，共享同一種對變化的感受，並因而被觸動去採取行動。換句話說，溝通就像一道波，穿過人們的身體，並在每個人都讓自己跟這道波同步時達到高峰。因此，場域參與者之間「心理與生理節奏」的共享，可以成為「socialization（社會化）」的驅動力。

概念化 Conceptualization

Once mutual trust and a common implicit perspective have been formed through shared experience, the team needs to articulate the perspective through continuous dialogues. The dominant mode of knowledge conversion here is externalization. Theories of organizational learning have not given much attention to this process. Tacit "field-specific" perspectives are converted into explicit concepts that can be shared beyond the boundary of the team. Dialogue directly facilitates this process by activating externalization at individual levels.

► 中文譯文

一旦相互信任與一個共同的內隱觀點透過共享經驗形成了，團隊就需要透過持續的對話，把這個觀點清楚地表達出來。這裡主要的知識轉換模式，是「externalization（外化）」。組織學習的相關理論，一直沒有給這個過程太多關注。那些隱性的、「特定於某個場域的」觀點，會被轉換成能跨越團隊邊界、被更廣泛共享的顯性概念。對話會在個人層次上活化「外化」，從而直接促成這個過程。

Dialogue, in the form of face-to-face communication between persons, is a process in which one builds concepts in cooperation with others. It also provides the opportunity for one's hypothesis or assumption to be tested. As Markova and Foppa (1990) argue, social intercourse is one of the most powerful media for verifying one's own ideas. As such, dialogue has a congenetic quality, and thus the participants in the dialogue can engage in the mutual codevelopment of ideas. As Graumann (1990) points out, dialogue involves "perspective-setting, perspective-taking, and multiperspectivity of cognition." According to the theory of language action suggested by Austin (1962) and Searle (1969), illocutionary speech does not only involve a description of things and facts but the taking of action itself. The expression "language is behavior," therefore, implies that language is a socially creative activity and accordingly reveals the importance of the connection between language and reality created through dialogue.

► 中文譯文

對話以人與人面對面溝通的形式進行，是一個人跟他人合作、共同建構概念的過程。它同時也提供了讓一個人的假設或預設「接受檢驗」的機會。如同 Markova 與 Foppa (1990) 所主張的，社會交往是用來驗證自己想法的最有力的媒介之一。正因如此，對話具有一種「共生（congenetic）」的特質，讓參與對話的人能投入想法的「相互共同發展」。如同 Graumann (1990) 所指出的，對話牽涉到「觀點的設定、觀點的採納，以及認知的多重觀點性」。根據 Austin (1962) 與 Searle (1969) 所提出的語言行動理論，「以言行事（illocutionary）」的言說，不只牽涉到對事物與事實的描述，更牽涉到行動本身的執行。因此，「語言即行為」這個說法，意味著語言是一種具有社會創造性的活動，並因而揭示了「語言」與「透過對話所創造出來的真實」這兩者之間連結的重要性。

For these purposes, dialectic is a good way of raising the quality of dialogue. Dialectic allows scope for the articulation and development of personal theories and beliefs. Through the use of contradiction and paradox, dialectic can serve to stimulate creative thinking in the organization. If the creative function of dialectic is to be exploited to the full, it is helpful to pay regard to certain preconditions or "field rules." First, the dialogue should not be single-faceted and deterministic but temporary and multifaceted so that there is always room for revision or negation. Second, the participants in the dialogue should be able to express their own ideas freely and candidly. Third, negation for the sake of negation should be discouraged. Constructive criticism substantiated by reasoned arguments should be used to build a consensus. Fourth, there should be temporal continuity. Dialectic thinking is a repetitive, spiral process in which affirmation and negation are synthesized to form knowledge. Strict and noncontinuous separation of affirmation and negation will only result in logical contradictions and thus hamper the creation of knowledge. Team leaders, therefore, should not discourage the dramatic and volatile dimensions of dialogue. If these conditions are met, dialogue will add much to the potential of the group in knowledge creation.

► 中文譯文

為了達到這些目的，「辯證（dialectic）」是提升對話品質的好方法。辯證為個人理論與信念的表達和發展，留出了空間。透過運用矛盾與弔詭，辯證能用來激發組織裡的創造性思考。如果能把辯證的創造功能發揮到極致，去留意某些前提條件，也就是「場域規則」，會很有幫助。第一，對話不該是單一面向、一翻兩瞪眼的，而應該是暫時的、多面向的，好讓修正或否定永遠都有空間。第二，參與對話的人應該能自由、坦率地表達自己的想法。第三，應該勸阻那種「為否定而否定」。要用有理據支撐的建設性批評，來建立共識。第四，應該要有時間上的連續性。辯證式思考是一個重複的、螺旋式的過程，在這個過程裡，肯定與否定被綜合起來，形成知識。如果硬把肯定與否定做嚴格而不連續的切割，只會造成邏輯上的矛盾，從而妨礙知識的創造。因此，團

隊領導者不該去壓抑對話裡那些戲劇化、易變動的面向。只要這些條件都滿足了，對話就會大大提升群體在知識創造上的潛力。

The process of creating a new perspective through interpersonal interaction is assisted by the existence of a degree of redundant information. Making and solving new problems are made possible when its members share information by obtaining extra, redundant information which enables them to enter another person's area and give advice. Instances of "learning by intrusion" (Nonaka 1990) are particularly widespread in Japanese firms. In the meantime, redundancy of information also functions to determine the degree to which created perspectives are diffused. It may sound paradoxical; yet the degree of information redundancy will limit the degree of diffusion. In this sense, information redundancy can serve to regulate the creation of perspectives.

► 中文譯文

透過人際互動去創造新觀點的這個過程，會因為存在某種程度的「冗餘資訊」而獲得助力。當成員們透過取得額外的、冗餘的資訊來共享資訊，使得彼此能進入別人的領域、給出建議時，提出新問題與解決新問題才變得可能。「藉由侵入來學習（learning by intrusion）」（Nonaka 1990）的例子，在日本企業裡特別普遍。同時，資訊的冗餘也具有一個功能，就是決定「被創造出來的觀點」能擴散到什麼程度。這聽起來或許有點弔詭，但資訊冗餘的程度，反而會「限制」擴散的程度。從這個意義上說，資訊冗餘可以用來「調節」觀點的創造。

It is now possible to turn to the question of how to conceptualize new perspectives created from shared tacit knowledge. According to Bateson (1979), concepts are created through deduction, induction, and abduction. Abduction has a particular importance in the conceptualization process. While deduction and induction are vertically-oriented reasoning processes, abduction is a lateral extension of the reasoning process which centers on the use of metaphors. Deduction and induction are generally used when a thought or image involves the revision of a preexisting concept or the assigning of a new meaning to a concept. When there is no adequate expression of an image it is necessary to use abductive methods to create completely new concepts. While analytical methods can be used to generate new concepts via inductive or deductive reasoning, they may not be sufficient to create more meaningful—or radical—concepts. At the early stages of information creation, it is very useful to pursue creative dialogues and to share images through the metaphorical process by merging perspectives, i.e., tacit knowledge.

► 中文譯文

現在可以轉到另一個問題：要如何把「從共享隱性知識所創造出來的新觀點」加以概念化。根據 Bateson (1979)，概念是透過演繹、歸納與「溯因（abduction）」被創造出來的。其中，溯因在概念化的過程裡有特別的重要性。演繹與歸納是垂直導向的推理過程，溯因則是推理過程的一種橫向延伸，它以「隱喻的運用」為核心。當一個想法或意象牽涉到「修正既有概念」、或「賦予某個概念新意義」時，一般會用演繹與歸納。但當一個意象「找不到適當的表達方式」時，就必須用溯因的方法，去創造出全新的概念。雖然分析性的方法可以透過歸納或演繹推理產生新概念，但它們可能不足以創造出更有意義的、或更具突破性的概念。在資訊創造的早期階段，去進行創造性的對話、並透過融合各種觀點（也就是隱性知識）的這種隱喻過程來共享意象，是非常有用的。

3.3 結晶化 3.3 Crystallization

The knowledge created in an interactive field by members of a self-organizing team has to be crystallized into some concrete "form" such as a product or a system. The central mode of knowledge conversion at this stage is internalization. Crystallization may then be seen as the process through which various departments within the organization test the reality and applicability of the concept created by the self-organizing team. These internalization processes are facilitated by encouraging experimentation.

► 中文譯文

由自組織團隊的成員在一個互動場域裡所創造出來的知識，必須被「結晶化」成某種具體的「形

式」，例如一個產品或一套系統。在這個階段，核心的知識轉換模式是「internalization（內化）」。因此，結晶化可以被看成是這樣一個過程：組織內部各個不同的部門，去檢驗自組織團隊所創造出來的那個概念，在現實中到底站不站得住腳、能不能實際應用。而這些內化的過程，會因為鼓勵實驗而更容易發生。

It should be noted that because the instrumental skill, a part of tacit knowledge, is exploited in this process, a new process of knowledge creation is triggered by crystallization. While this usually leads to refinement of the concept, sometimes the concept itself is abandoned and fundamentally recreated.

► 中文譯文

值得注意的是，由於在這個過程裡，會用到「工具性技能」（這也是隱性知識的一部分），所以結晶化會觸發一個新的知識創造過程。雖然這通常會讓概念變得更精煉，但有時候，概念本身會被整個放棄，然後從根本上重新創造。

The process of crystallization is a social process which occurs at a collective level. It is realized through what Haken (1978) called "dynamic cooperative relations" or "synergetics" among various functions and organizational departments. This relationship tends to be achieved most effectively when redundancy of information creates scope for critical knowledge conversion processes to take place. In an organization where there is redundancy of information, the initiative for action can be taken by the experts who have more information and knowledge. This characteristic is what McCulloch (1965) called "the principle of redundancy of potential command." In this principle, all parts of a system carry the same degree of importance, and each part's impact upon the system is determined by the importance of information it contains in each specified context. In sum, each part has the potential of becoming the leader of the entire system when there exists redundancy of information.

► 中文譯文

結晶化的過程，是一個發生在集體層次的社會過程。它是透過 Haken（1978）所說的、各種功能與組織部門之間的「動態合作關係」或「協同學（synergetics）」來實現的。當資訊的冗餘為那些關鍵的知識轉換過程創造出空間時，這種關係往往最能有效達成。在一個資訊有冗餘的組織裡，採取行動的主導權，可以由那些掌握較多資訊與知識的專家來拿。這個特性，正是 McCulloch（1965）所說的「潛在指揮權的冗餘原則」。在這個原則裡，系統的每一個部分都帶有同等程度的重要性，而每個部分對系統的影響，則取決於它在每個特定脈絡裡所含資訊的重要性。總的來說，當資訊冗餘存在時，每一個部分都有潛力成為整個系統的領導者。

The speed at which Japanese firms develop new products seems to be assisted by information redundancy. In the product development process of Japanese firms, different phases of the process are loosely linked, overlapping in part, and the creation and realization of information is carried out flexibly. The loosely linked phases, while simultaneously maintaining mutual independence, have redundant information that activates their interactive inquiry thereby facilitating cyclical generation and solution of problems (Imai et al. 1985). This "rugby-style" product development is equipped with the flexible capability of knowledge conversion. Clark and Fujimoto (1991) showed that Japanese firms take relatively less time for product development than American and European firms.

► 中文譯文

日本企業開發新產品的速度，看來是受惠於資訊冗餘。在日本企業的產品開發過程裡，過程中不同的階段是「鬆散地連結」在一起的，彼此部分重疊，而且資訊的創造與實現是靈活進行的。這些鬆散連結的階段，一方面維持著彼此的相互獨立，一方面又擁有冗餘的資訊，去活化它們之間互動式的探問，從而促成問題的循環式產生與解決（Imai 等人 1985）。這種「橄欖球式」的產品開發，配備了靈活的知識轉換能力。Clark 與 Fujimoto（1991）的研究顯示，比起美國與歐洲企業，日本企業在產品開發上花的時間相對較少。

The specific characteristics of the product development in Japanese firms is its lateral breadth covering the whole organization. In other words it is overlapping and synthetic rather than analytic or linear. In this system, development staff can traverse overlapping phases and, to a certain extent, share each other's functions. This is far different from the

usual product development process of U.S. firms, which have definite partitions between phases over which a baton is relayed. In the Japanese "rugby-style" product development (Takeuchi and Nonaka 1986), staff involved in one phase also may be in the next phase. Thus, some development staff can be involved in all phases of development. Sometimes this process also involves those outside the organization such as suppliers and customers in order to mobilize and explicit environmental knowledge.

► 中文譯文

日本企業產品開發的獨特之處，在於它那種橫向、涵蓋整個組織的廣度。換句話說，它是重疊的、綜合性的，而不是分析式的或線性的。在這套系統裡，開發人員可以橫跨彼此重疊的各個階段，並在某種程度上分擔彼此的功能。這跟美國企業一般的產品開發過程很不一樣，後者各階段之間有明確的分界，工作像接力棒一樣一棒一棒往下傳。在日本的「橄欖球式」產品開發裡 (Takeuchi and Nonaka 1986)，參與某一個階段的人員，也可能身處在下一個階段。因此，有些開發人員可以參與開發的所有階段。有時候，這個過程也會把組織外部的人，例如供應商與顧客，一起拉進來，以便動員環境裡的知識、並把它顯性化。

One problem with this developmental style is the potential risk of confusion if, for example, the design changes or other alterations take place. Participants might have to exert more effort to organize the process due to the lack of strict specifications at each phase and definite boundaries between them. However, these risks are counterbalanced by a tendency to create and realize concepts quickly and flexibly in an integral fashion. In this context, redundant information can play a major role in facilitating the process.

► 中文譯文

這種開發風格有一個問題，就是萬一發生像是設計變更之類的更動，可能會有造成混亂的潛在風險。由於每個階段缺乏嚴格的規格、階段之間也沒有明確的界線，參與者可能必須花更多力氣去把過程組織起來。不過，這些風險被另一種傾向所平衡掉了，那就是能夠以一種整合的方式，又快又靈活地創造並實現概念。在這個脈絡下，冗餘資訊可以在促進這個過程上扮演重要角色。

3.4 知識的證成與品質 3.4 The Justification and Quality of Knowledge

While organizational knowledge creation is a continuous process with no ultimate end, an organization needs to converge this process at some point in order to accelerate the sharing of created knowledge beyond the boundary of the organization for further knowledge creation. As knowledge is conventionally defined as "justified true belief," this convergence needs to be based on the "justification" or truthfulness of concepts. Justification is the process of final convergence and screening, which determines the extent to which the knowledge created within the organization is truly worthwhile for the organization and society. In this sense, justification determines the "quality" of the created knowledge and involves criteria or "standards" for judging truthfulness.

► 中文譯文

雖然組織知識創造是一個沒有最終終點的連續過程，但組織需要在某個時間點讓這個過程收斂，好加速把創造出來的知識分享出去，跨越組織的邊界，以進行進一步的知識創造。由於知識按慣例被定義為「經過證成的真實信念 (justified true belief)」，這個收斂必須建立在概念的「證成 (justification)」或「真確性」之上。證成，就是最後收斂與篩選的過程，它決定了組織內部所創造出來的知識，對組織與社會來說，到底有多大的真正價值。從這個意義上說，證成決定了所創造知識的「品質」，並牽涉到用來判斷真確性的判準或「標準」。

What matters here are the evaluation "standards" for judging truthfulness. In the business organizations, the standards generally include cost, profit margin, and the degree to which a product can contribute to the firm's development. There are also value premises that transcend factual or pragmatic considerations. These might be opinions about such things as the extent to which the knowledge created is consistent with the organization's vision and perceptions relating to adventure, romanticism, and aesthetics. The inducements to initiate a convergence of knowledge may be multiple and qualitative rather than simple and quantitative standards such as efficiency, cost, and return on investment (ROI).

► 中文譯文

這裡真正關鍵的，是用來判斷真確性的評估「標準」。在企業組織裡，這些標準一般包括成本、利潤率，以及一個產品能對公司發展貢獻多少。此外還有一些超越事實面或實用面考量的「價值前提」。這些可能是一些看法，例如所創造的知識跟組織願景一致到什麼程度，以及跟冒險、浪漫情懷、美學有關的種種知覺。促使知識開始收斂的誘因，可能是多重而質性的，而不是像效率、成本、投資報酬率（ROI）那樣單純而量化的標準。

In knowledge-creating organizations, it is the role of top or middle management to determine the evaluation standard. Determining the turning point from dissipation to convergence in the creation process is a highly strategic task which is influenced by the "aspiration" of the leaders of the organization. Justification standards have to be evaluated in terms of their consistency with higher-order value systems. The ability of leaders to maintain continuous self-reflection in a wider perspective is indispensable when it comes to increasing the quality of knowledge created.

► 中文譯文

在以知識創造為核心的組織裡，決定評估標準是高階或中階管理者的角色。要決定創造過程裡從「耗散」轉向「收斂」的那個轉捩點，是一項高度策略性的任務，它會受到組織領導者的「企圖心（aspiration）」所影響。證成的標準，必須以它們跟「更高層次價值體系」是否一致來加以評估。當談到要提升所創造知識的品質時，領導者那種在更寬廣的視野裡持續進行自我反思的能力，是不可或缺的。

3.5 知識的網絡化 3.5 Networking Knowledge

The realization of new concepts, described above, represents a visible emergence of the organization's knowledge network. During this stage of organizational knowledge creation, the concept that has been created, crystallized and justified in the organization is integrated into the organizational knowledge-base which comprises a whole network of organizational knowledge. The organizational knowledge base is then reorganized through a mutually-inducing process of interaction between the established organizational vision and the newly-created concept.

► 中文譯文

前面所描述的新概念的實現，代表了組織知識網絡一次看得見的浮現。在組織知識創造的這個階段裡，那個已經在組織內被創造、被結晶化、被證成的概念，會被整合進「組織知識庫」，而這個知識庫構成了一整張組織知識的網絡。接著，組織知識庫會透過一個「相互誘發」的過程被重新組織，這個過程，就是既有的組織願景與新創造出來的概念之間的互動。

Speaking in sociological terms, this mutually-inducing relationship corresponds to the relationship between a grand concept and a middle-range concept. A middle-range concept is induced from an equivocal knowledge base as a grand concept and then is condensed into concrete form. The grand concept is not fully understood at the organizational level unless these middle-range concepts are verified on site. This verification also induces the creation or reconstruction of a grand concept, causing the interactive proliferation of grand concepts presented by top management, and middle-range concepts created by middle management. This interaction, mediated by the concrete form as condensed information, is another dynamic self-organizing activity of knowledge network that continuously creates new information and meaning.

► 中文譯文

用社會學的話來說，這種相互誘發的關係，對應的是「宏大概念（grand concept）」與「中程概念（middle-range concept）」之間的關係。中程概念是從一個如同宏大概念那般「模稜兩可」的知識庫裡被誘發出來，然後再被濃縮成具體的形式。除非這些中程概念在現場被驗證，否則宏大概念在組織層次上不會被完全理解。而這種驗證，又會反過來誘發宏大概念的創造或重建，造成高階管理者所提出的宏大概念、與中階管理者所創造的中程概念之間，互動式地不斷增生。這種互動，以被濃縮成資訊的「具體形式」為中介，是知識網絡另一種動態的自組織活動，會持續創造出新的資訊與意義。

It should be noted that the process of organizational knowledge creation is a never-ending, circular process that is not confined to the organization but includes many interfaces with the environment. At the same time, the environment is a continual source of stimulation to knowledge creation within the organization. For example, Hayek (1945) pointed out that the essential function of market competition is to discover and mobilize knowledge "on-the-spot," i.e., the implicit, context-specific knowledge held by market participants.

► 中文譯文

值得注意的是，組織知識創造的過程，是一個永不止息的、循環的過程，它不侷限在組織之內，而是包含許多跟環境之間的介面。同時，環境也是組織內部知識創造源源不絕的刺激來源。舉例來說，Hayek（海耶克，1945）指出，市場競爭最根本的功能，就是去發現並動員那種「就在現場（on-the-spot）」的知識，也就是市場參與者所持有的、隱性的、特定於脈絡的知識。

In the case of business organizations, one aspect of the relationship between knowledge creation and the environment is illustrated by reactions to the product by customers, competitors, and suppliers. For example, many dimensions of customer needs take the form of tacit knowledge that an individual customer or other market participants cannot articulate by themselves. **A product works as a trigger to articulate the tacit knowledge.** Customers and other market participants give meaning to the product by their bodily actions of purchasing, adapting, using, or not purchasing. This mobilization of tacit knowledge of customers and market will be reflected to the organization, and a new process of organizational knowledge creation is again initiated.

► 中文譯文

以企業組織來說，知識創造與環境之間關係的其中一個面向，可以從顧客、競爭者與供應商對產品的反應裡看出來。舉例來說，顧客需求的許多面向，是以隱性知識的形式存在的，個別顧客或其他市場參與者自己沒辦法把它清楚說出來。產品則扮演一個「觸發器」的角色，去把這些隱性知識表達出來。顧客與其他市場參與者，透過他們購買、調整、使用、或不購買等等的身體行動，賦予了產品意義。這種對顧客與市場隱性知識的動員，會回饋到組織裡，於是一個新的組織知識創造過程，又再次被啟動。

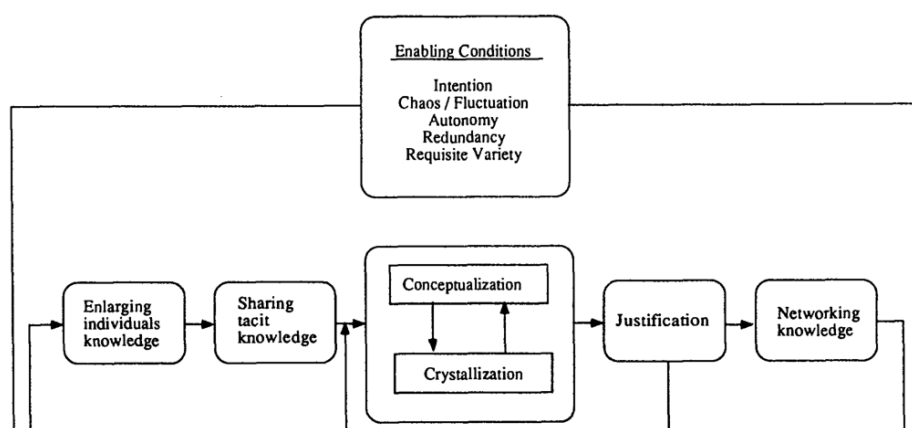


Figure 3. Organizational Knowledge Creation Process

圖 3 組織知識創造過程

The total process of organizational knowledge creation is summarized in Figure 3. Even though the figure is illustrated as a sequential model, the actual process progresses forming multilayered loops. Respective stages can take place simultaneously, or sometimes jump forward or backward.

► 中文譯文

組織知識創造的整個過程，總結在圖 3 裡。儘管這張圖被畫成一個「依序進行」的模型，但實際的過程是以「多層次的迴圈」往前推進的。各個階段可以同時發生，有時候也會往前跳、或往後退。